

Does the Hubbard Model Secretly Contain a Closed Universe?

Brian Swingle (Brandeis)
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With [Valerie Bettaque](#) (2602.12339),
[Alejandro Vilar Lopez](#) and [Martin Sasieta](#) (2512.00149), and [Shiyong Guo](#) (WIP)



Alchemical ouroboros ([Wikipedia](#))



Valerie Bettaque



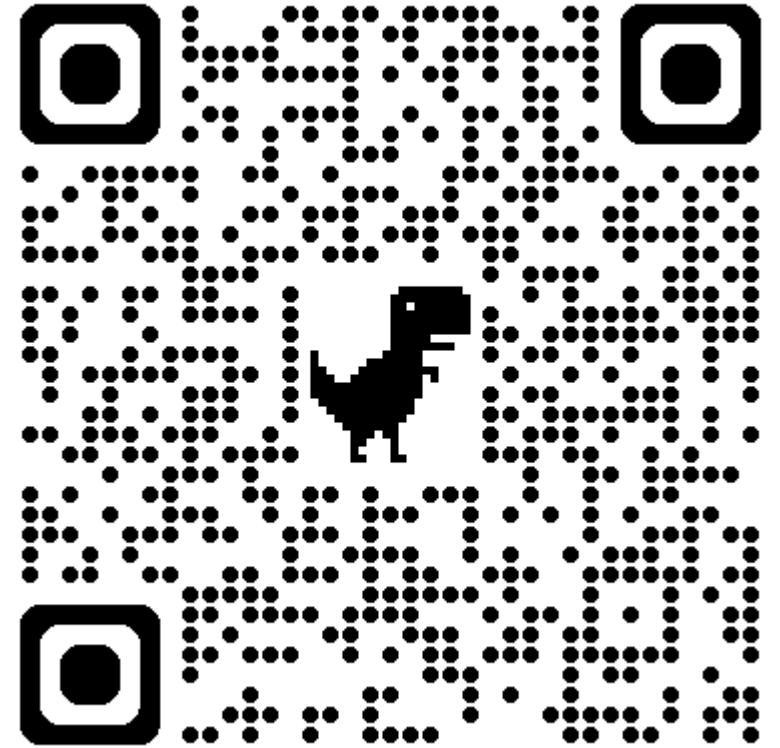
Shiyong Guo



Alejandro Vilar Lopez



Martin Sasieta



physicsmonkey.github.io

+ AI systems (Opus,
Fable, Sol, Astra)

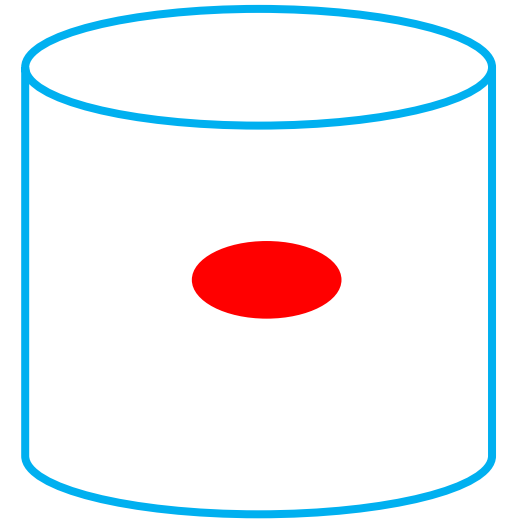
Plan

- Setting the stage ([why is this a condensed matter talk???](#))
 - Motivation from wormholes and closed universes
 - Non-stabilizerness aka magic
 - Quantum advantage?
- The SYK computation
- Wormholes in SYK
- Magic in SYK
- Advantage in SYK?
- Generalizing, e.g. Hubbard?

AdS/CFT and “it from qubit”

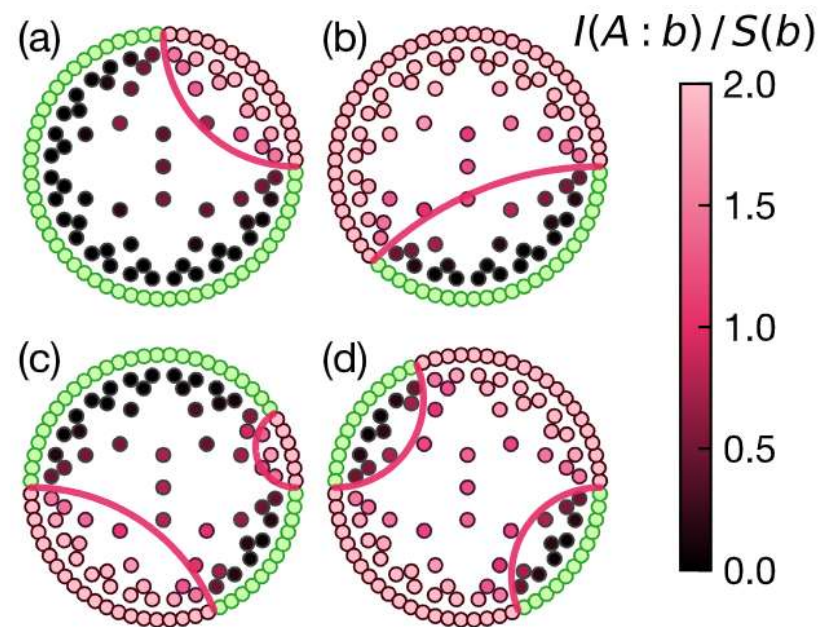
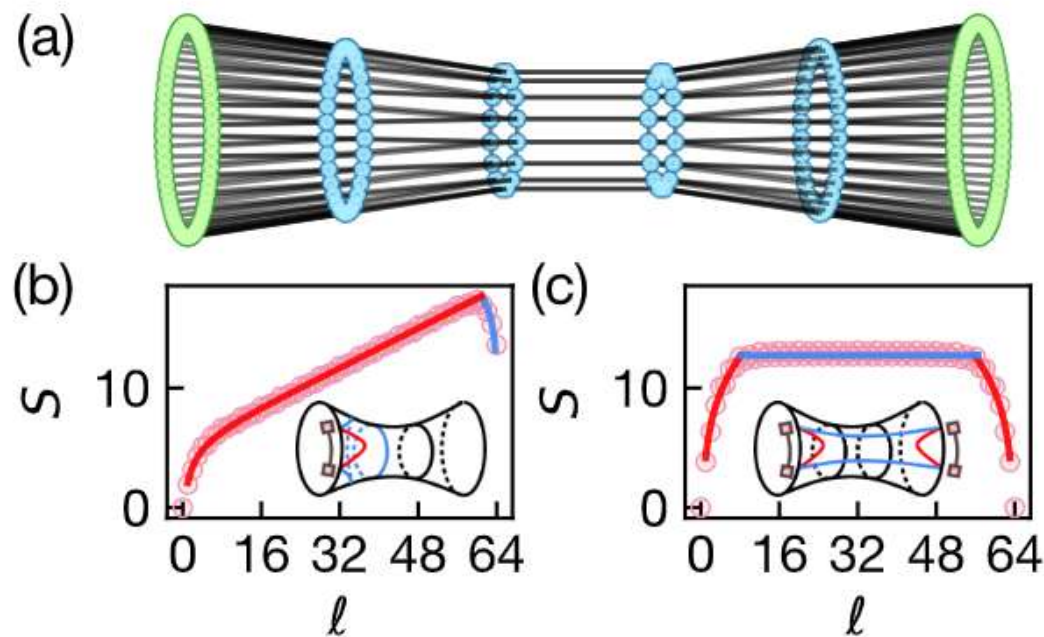
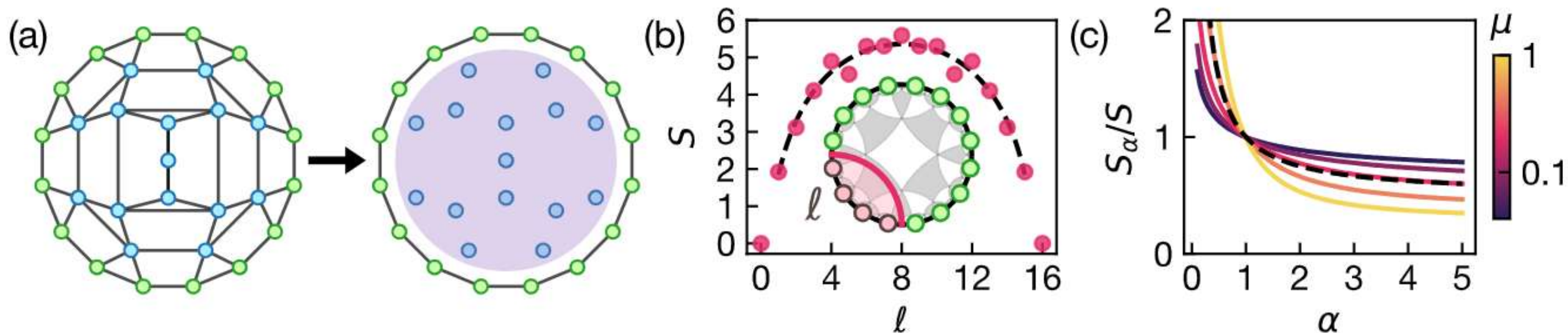
For a “universe in a bottle” (Anti de Sitter space), the AdS/CFT correspondence gives a framework for quantum gravity: [Maldacena, ...]

- Wonderful theoretical lab to study black holes, strong coupling dynamics, hydrodynamics, ...
- Gravitational physics is intimately linked with quantum information: entanglement and geometry, complexity and black holes, ...
- By simulating the boundary system in a lab, we can hope to experimentally study holography!



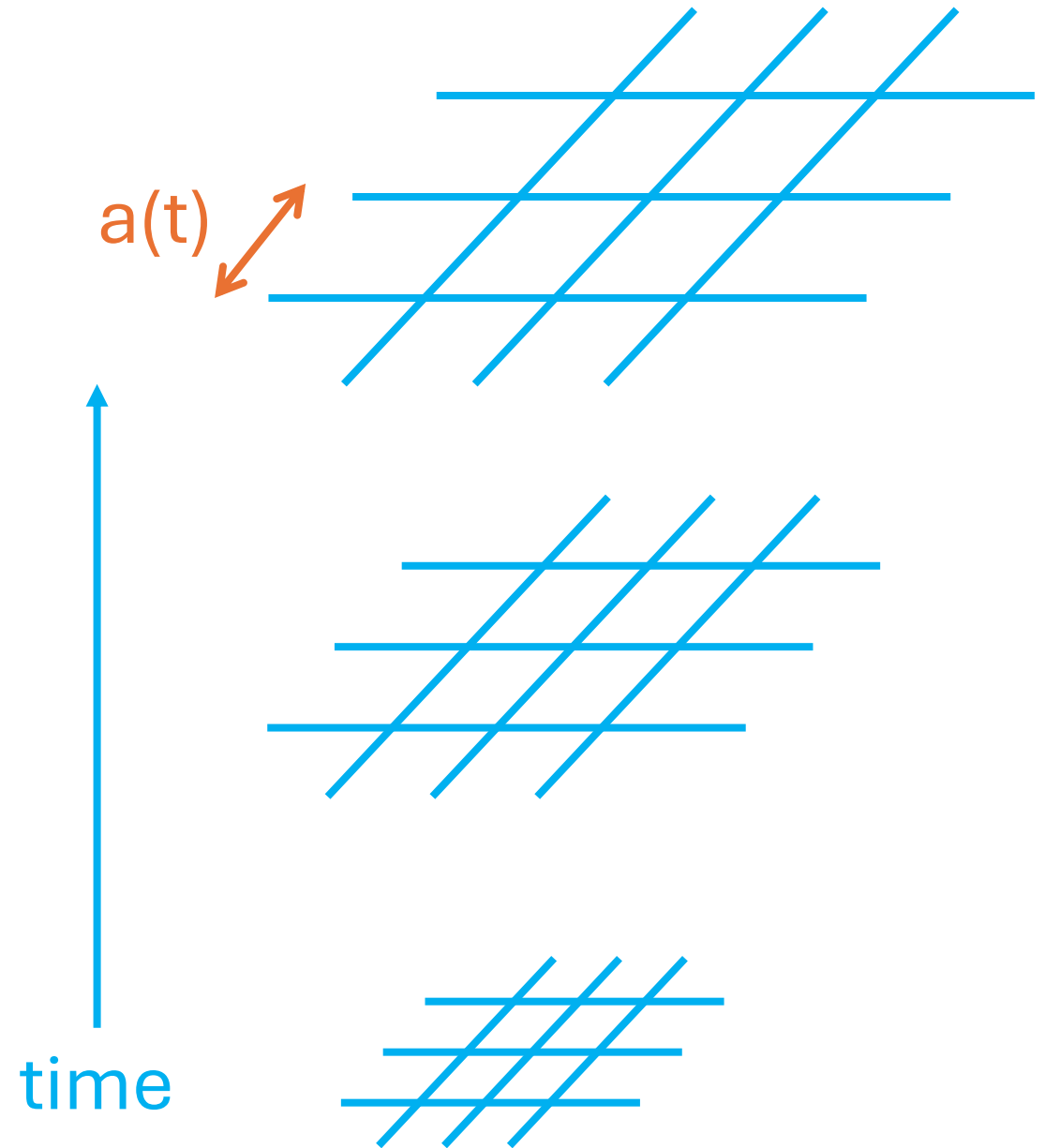
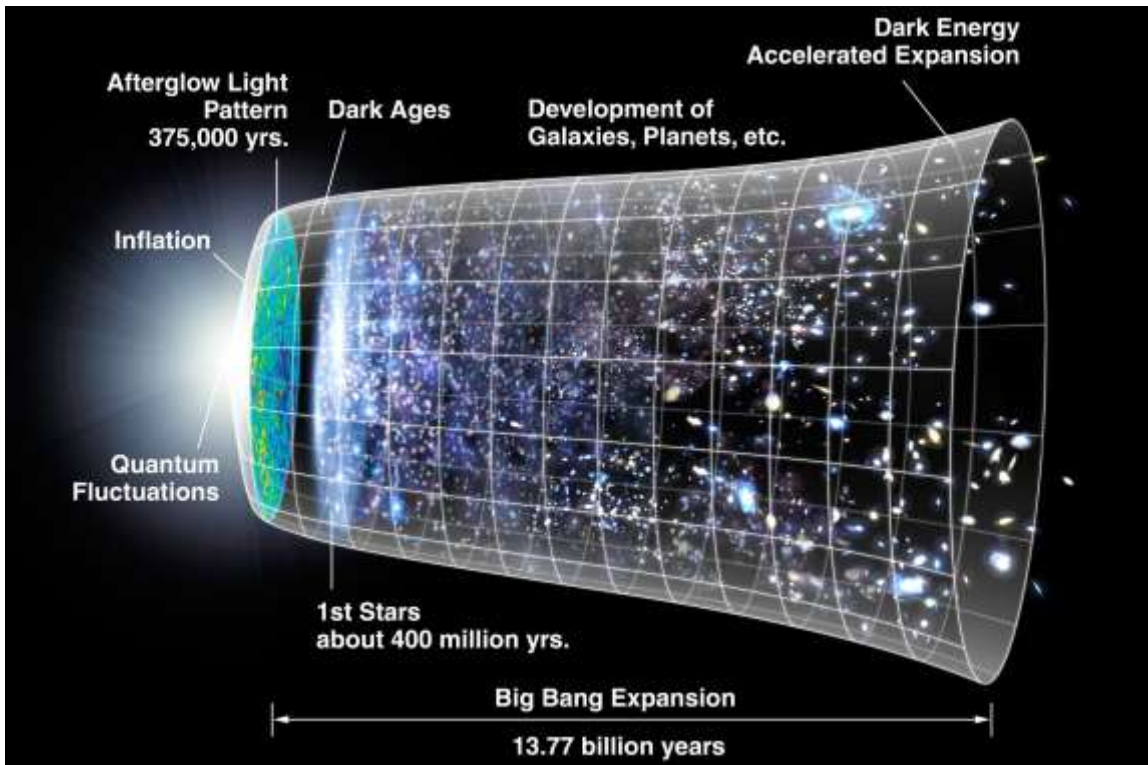
Stuff in AdS $\leftrightarrow$
States in dual CFT

["Building Holographic Entanglement by Measurement",
Jeffrey-Gandarias-Schleier-Smith-S 2601.21065]



But our universe is:

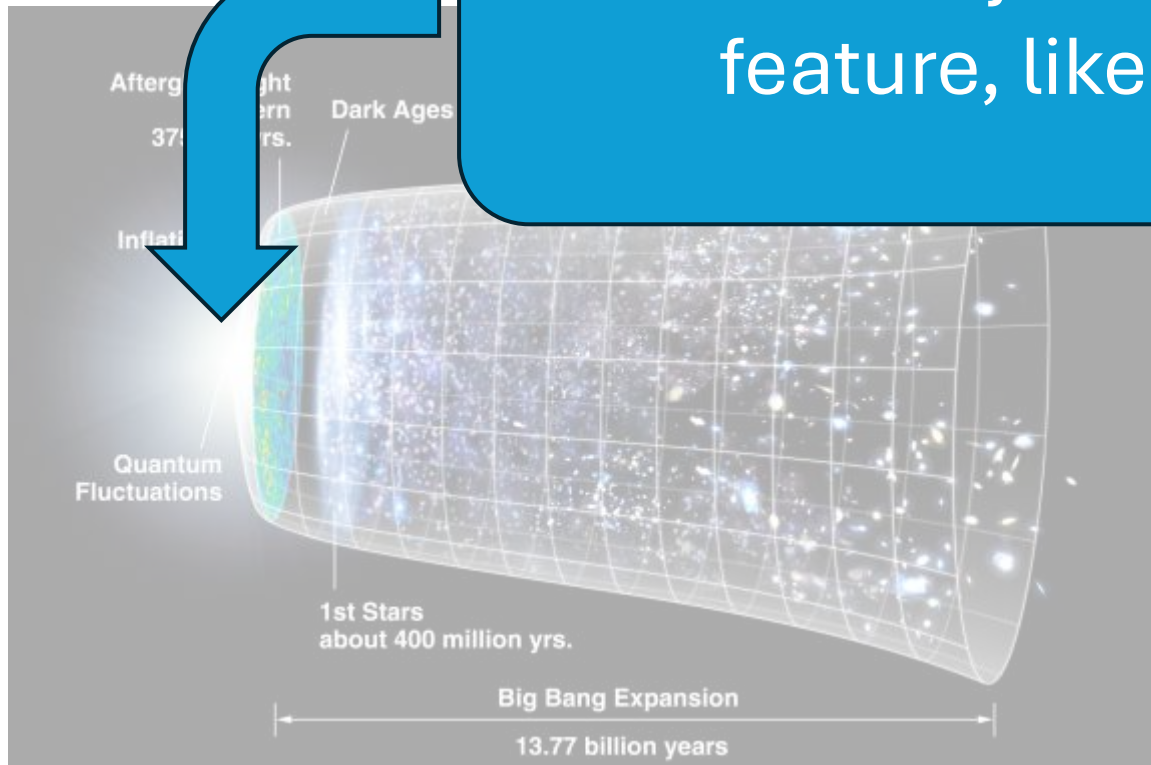
- Full of stuff, homogenous, ...
- Expanding: $a(t)$
- Λ CDM: positive dark energy
- **No AdS boundaries**



But our universe is:

- Full of stuff, homogenous, ...
- Expanding: $a(t)$
- Λ CDM: positive dark energy
- No AdS boundary

What if we just wanted a single feature, like a Big Bang?



time

$a(t)$

A toy Big Bang!

Flat + radiation + $\Lambda < 0$
(negative dark energy as in AdS/CFT)

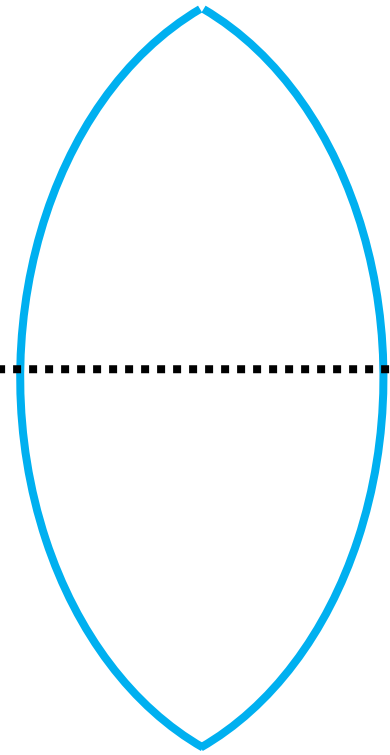
$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2)$$

$$a(t)^2 = \cos t/t_*$$

symmetric



Big Crunch



Big Bang

No AdS boundaries; this is a
flat and homogeneous space

time

Towards realism?

Flat + radiation + $\Lambda < 0$

$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2)$$

$$a(t)^2 = \cos t/t_*$$

symmetric



Big Crunch



Add enough bells and whistles (e.g. scalar fields) → you are here??

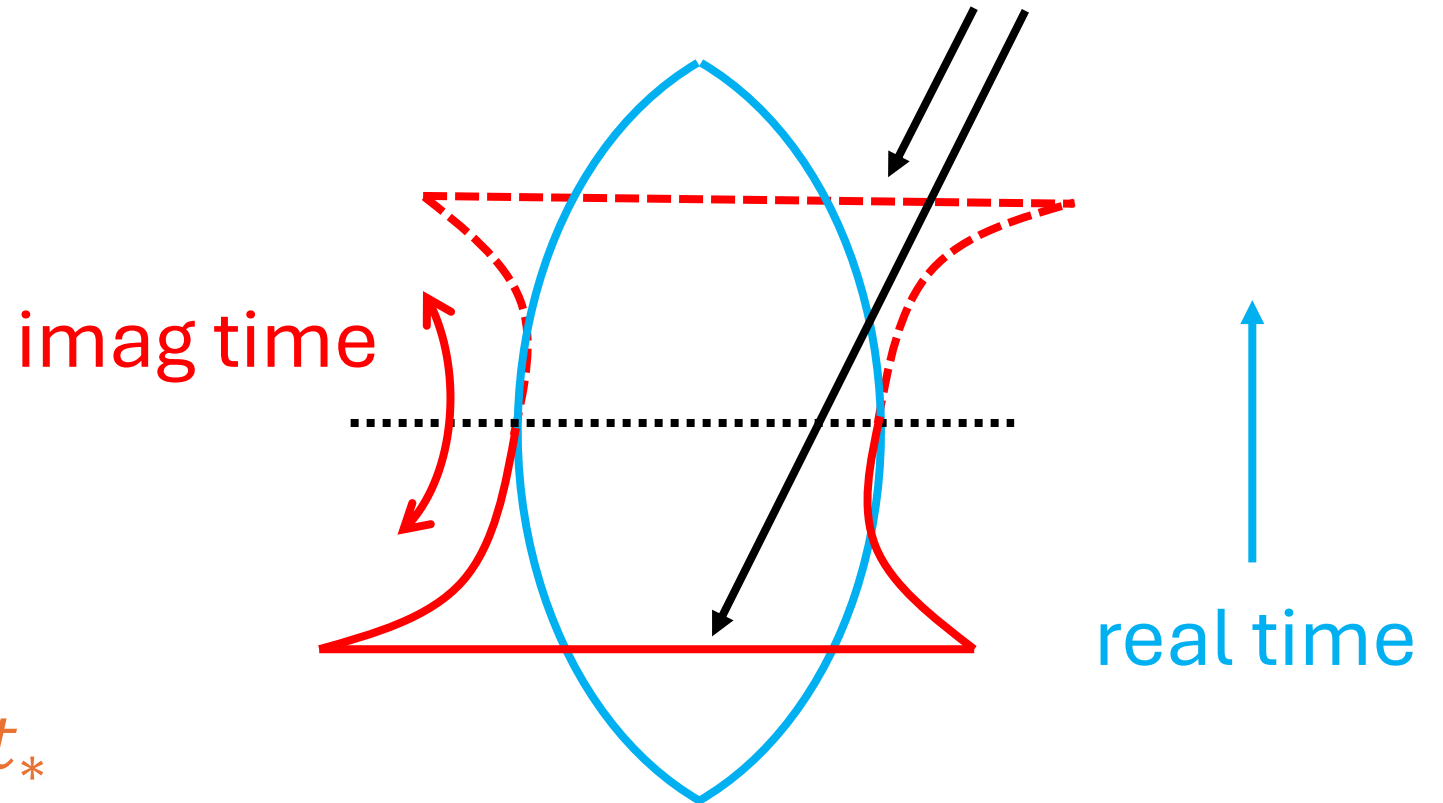
Big Bang

time

[more realistic models: Antonini-Waddell-Simidzija-Van Raamsdonk-S, ...]

Meaning?

Euclidean wormhole connecting
two distinct AdS boundaries

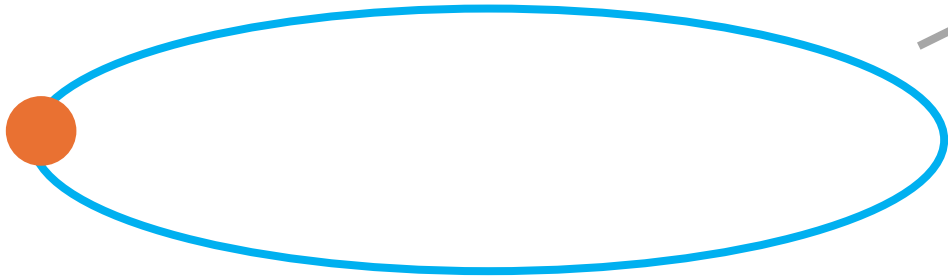


$$a(t)^2 = \cos t/t_*$$

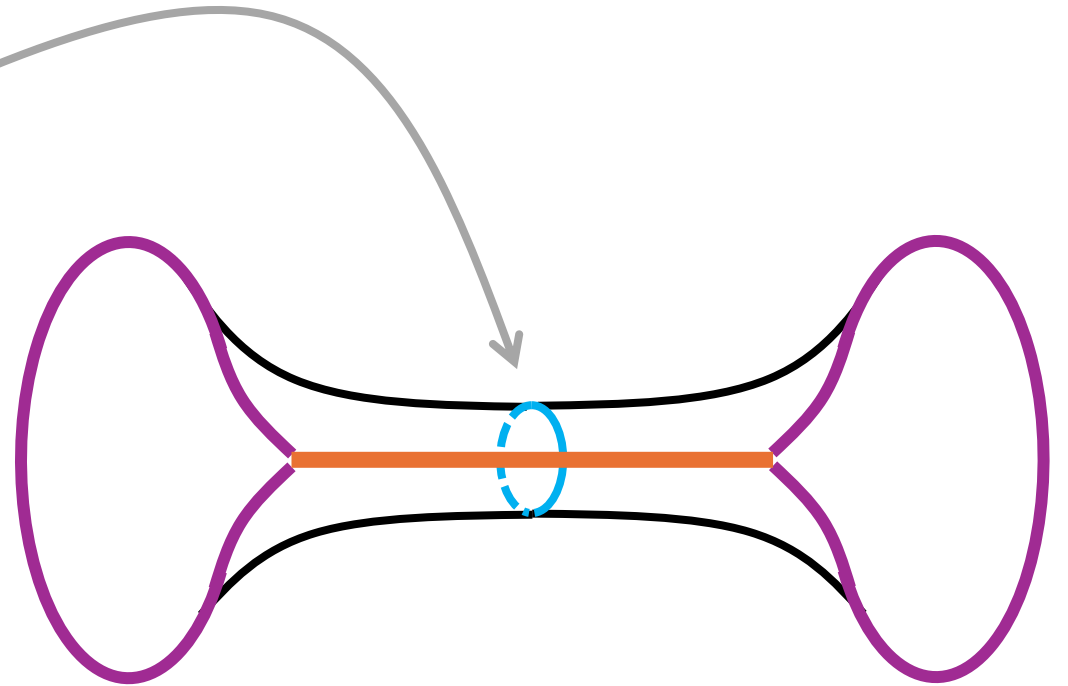
$$t \rightarrow i\tau: a(\tau)^2 = \cosh \tau/t_*$$

[Maldacena-Maoz, McInnes, Antonini-
Waddle-Simidzija-Van Raamsdonk-S, ...]

A toy model of the toy model



The space of the closed universe consists of a mostly empty **circle** with a **heavy clump of matter**



Euclidean wormhole with two **asymptotic regions**

Semi-classical bulk picture

JT gravity coupled to matter, action: $I = I_{\text{JT}} + I_{\text{matter}}$

$$I_{\text{JT}} = -S_0 \chi[\Sigma] - \frac{1}{2} \int_{\Sigma} \Phi (R + 2) - \int_{\partial \Sigma} \Phi_b (K - 1)$$

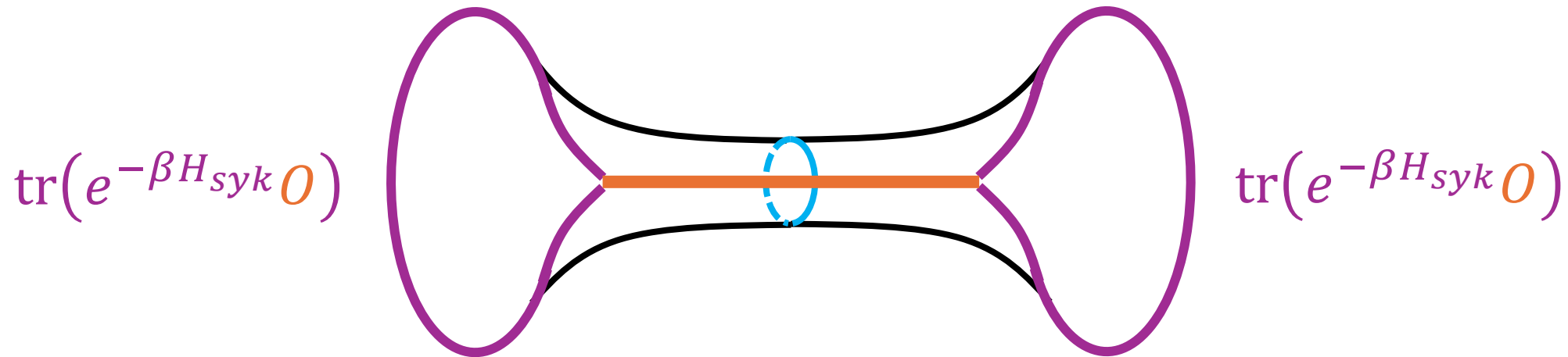
Field content:

- The metric is fixed to be constant curvature,
- The dilaton (Φ) can have a non-trivial profile
- The matter can be taken to be $O(N)$ massive fermion fields

[[Kitaev](#), [Sachdev](#), [Maldacena-Stanford](#), [Rosenhaus-Polchinski](#),...]

Microscopic connection

- The Euclidean wormhole gives a contribution to a statistical second moment (two boundaries) of erratic data in the dual theory
- We will make this precise using the Sachdev-Ye-Kitaev model



[connection to averaging: Coleman, Saad-Shenker-Stanford (SFF), Antonini-Waddell-Simidzija-Van Raamsdonk-S (cosmological), Stanford (heavy op), ...]

Consider a system of N Majorana fermions χ_i

Majorana strings, a complete basis of Hermitian operators: $\mu(a) \propto \chi_1^{a_1} \cdots \chi_N^{a_N}$ Weight = $\sum a_i$

Thermal 1-point functions, these completely determine the state: $\xi(a) = \frac{\text{tr} \left(e^{-\beta H} \mu(a) \right)}{\text{tr} \left(e^{-\beta H} \right)}$

We will consider ensembles of Hamiltonians and investigate the statistical properties of these 1-point functions

Sachdev-Ye-Kitaev model ($q=4$)

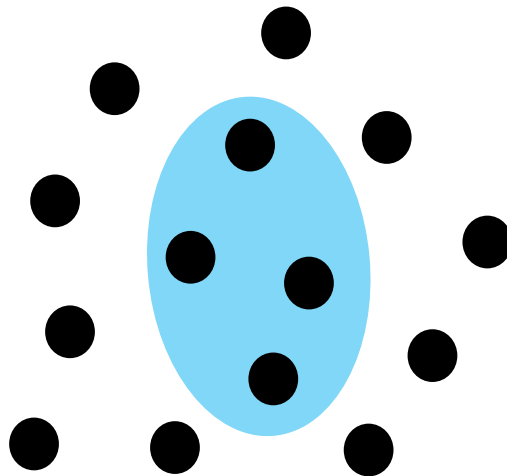
$$\chi_a^\dagger = \chi_a, \quad a = 1, \dots, N$$

$$H = \frac{1}{4!} \sum_{abcd} J_{abcd} \chi_a \chi_b \chi_c \chi_d$$

$$\{\chi_a, \chi_b\} = \delta_{a,b}$$

$$\overline{J_{abcd}} = 0, \quad \overline{J_{abcd}^2} = \frac{3! J^2}{N^3}$$

$$D = 2^{N/2}$$

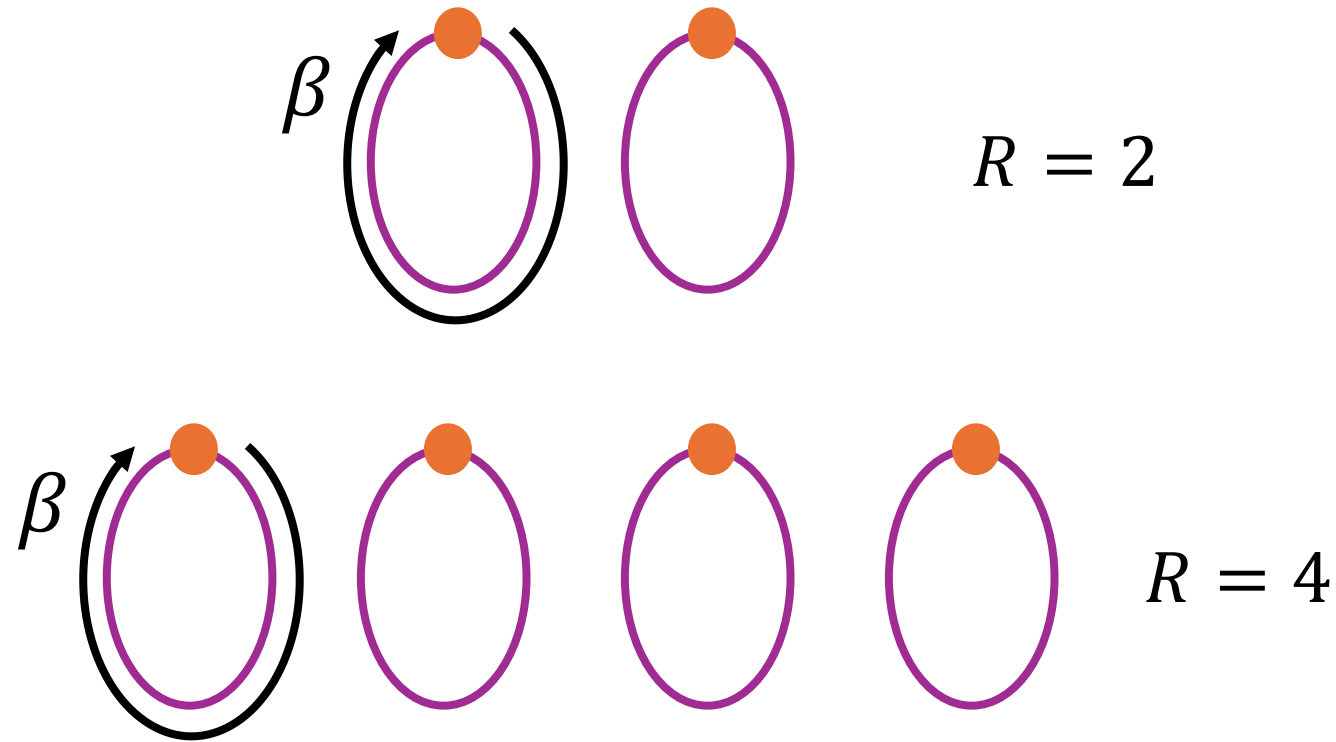


[random 2-body ensemble, Sachdev-Ye, Kitaev, Rosenhaus-Polchinski, Maldacena-Stanford, Garcia-Garcia-Verbaarschot, Bagrets-Altland-Kamenev, ...]

Statistical moments

$$\xi(a) = \frac{\text{tr} \left(e^{-\beta H} \mu(a) \right)}{\text{tr} \left(e^{-\beta H} \right)}$$

- We will compute the ensemble average of the R-th moment of the **thermal 1-point function** with a path integral on R replicas



1-point functions are distinct from ETH

- Eigenstate thermalization hypothesis (ETH):

$$\langle E_i | \mathcal{O} | E_j \rangle = \mathcal{O}(\epsilon) \delta_{ij} + e^{-S(\epsilon)/2} f_{\mathcal{O}}(\epsilon, \omega) R_{ij}.$$

- In contrast, we focus on diagonal matrix elements in ground states or thermal states and these may or may not be “erratic”

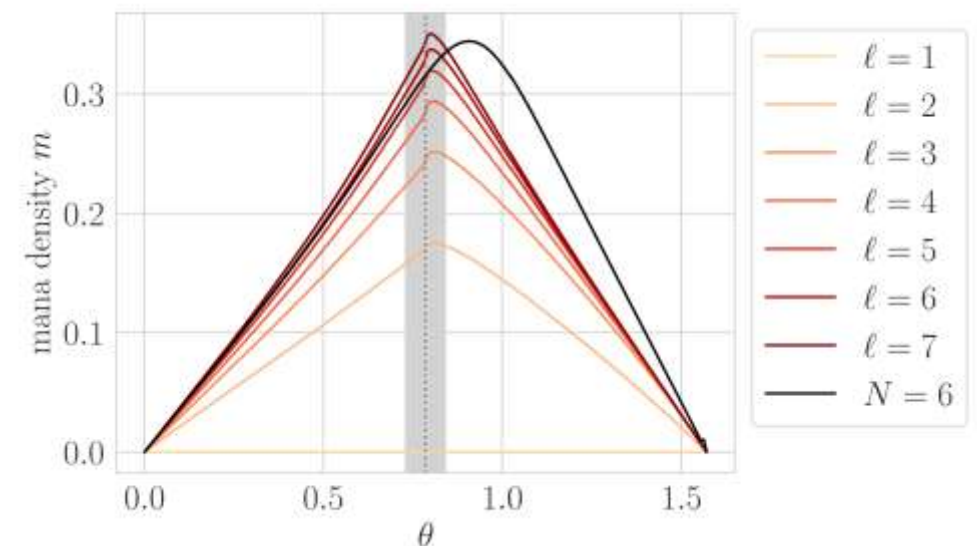
$$\xi(a) = \frac{\text{tr} \left(e^{-\beta H} \mu(a) \right)}{\text{tr} \left(e^{-\beta H} \right)}$$

1-point functions $\rightarrow$ non-stabilizerness

- Entanglement $\rightarrow$ how close is a state to being a product state?, non-Gaussianity $\rightarrow$ how close is a state to being non-interacting
- Magic/~~STAB~~ $\rightarrow$ how close is a state to being a ground state of a commuting projector Hamiltonian (there is some fine print here)
- This perspective has both computational and conceptual uses
- Early example (3 state Potts):

$$H = -\sin \theta \sum_j [Z_j^\dagger Z_{j+1} + h.c.] - \cos \theta \sum_j [X_j^\dagger + X_j]$$

[Cao-White-S 2007.01303]



1-point functions $\rightarrow$ quantum advantage?

- In the context of quantum simulation and computation, one goal is to find (scientifically interesting) problems that are plausibly hard to solve classically while being easy to solve quantumly
- SYK may seem like a strange place to look, e.g. use ED at small N and semi-classical methods at large N and patch them together?
- However, these 1-point functions are erratic in a single instance and zero on average, so since SYK is highly {entangled, complex, **magical**, ...} maybe these are hard to obtain classically?
- Nevertheless, we will argue that these low-weight 1-point functions can be efficiently computed in SYK

[S+Fable+Sol+Astra coming soon, see also Zlokapa-Kiani]

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- Generalizing, e.g. Hubbard?

Path integral computation

$$\begin{aligned}\mathrm{tr} \left(\mu(a) e^{-\beta H_{\mathrm{SYK}}} \right)^R &= \mathrm{tr} \left[\mu^{\otimes_f R} \exp(-\beta H_{\mathrm{SYK}})^{\otimes_f R} \right] \\ &= \mathrm{tr} \left[\mu^{(1)}(a) \mu^{(2)}(a) \cdots \mu^{(R)}(a) \exp \left(-\beta \sum_{r=1}^R H_{\mathrm{SYK}}^{(r)} \right) \right] \\ &\simeq \mathrm{tr} \left[(-1)^{a_1 F_{R,1}} \cdots (-1)^{a_N F_{R,N}} \exp \left(-\beta \sum_{r=1}^R H_{\mathrm{SYK}}^{(r)} \right) \right]\end{aligned}$$

$$(-1)^{F_{R,i}} := (2i)^{R/2} \chi_i^{(1)} \cdots \chi_i^{(R)}$$

Path integral computation

$$G_{rs}(\tau, \tau') := \frac{1}{N} \sum_{i=1}^N \chi_i^{(r)}(\tau) \chi_i^{(s)}(\tau'), \quad G_{rs}(\tau, \tau') = -G_{sr}(\tau', \tau),$$

multi-replica collective field

$$\overline{\text{tr}(\mu(a) e^{-\beta H_{\text{SYK}}})^R}$$

$W = \text{weight} = \# \text{ of non-identity ops in } \mu$

$$= \int [\mathcal{D}\Sigma] [\mathcal{D}G] \text{pf}[I_R \otimes \mathcal{D}^- - \Sigma]^{N-W} \text{pf}[I_R \otimes \mathcal{D}^+ - \Sigma]^W$$

$$\exp \left(-\frac{N}{2} \sum_{r,s=1}^R \int_0^\beta d\tau d\tau' \left(\Sigma_{rs}(\tau, \tau') G_{rs}(\tau, \tau') - \frac{J^2}{q} G_{rs}(\tau, \tau')^q \right) \right)$$

ensemble average washes out the dependence
on which fermions were chosen for μ

[Bettaque-S]

Path integral computation

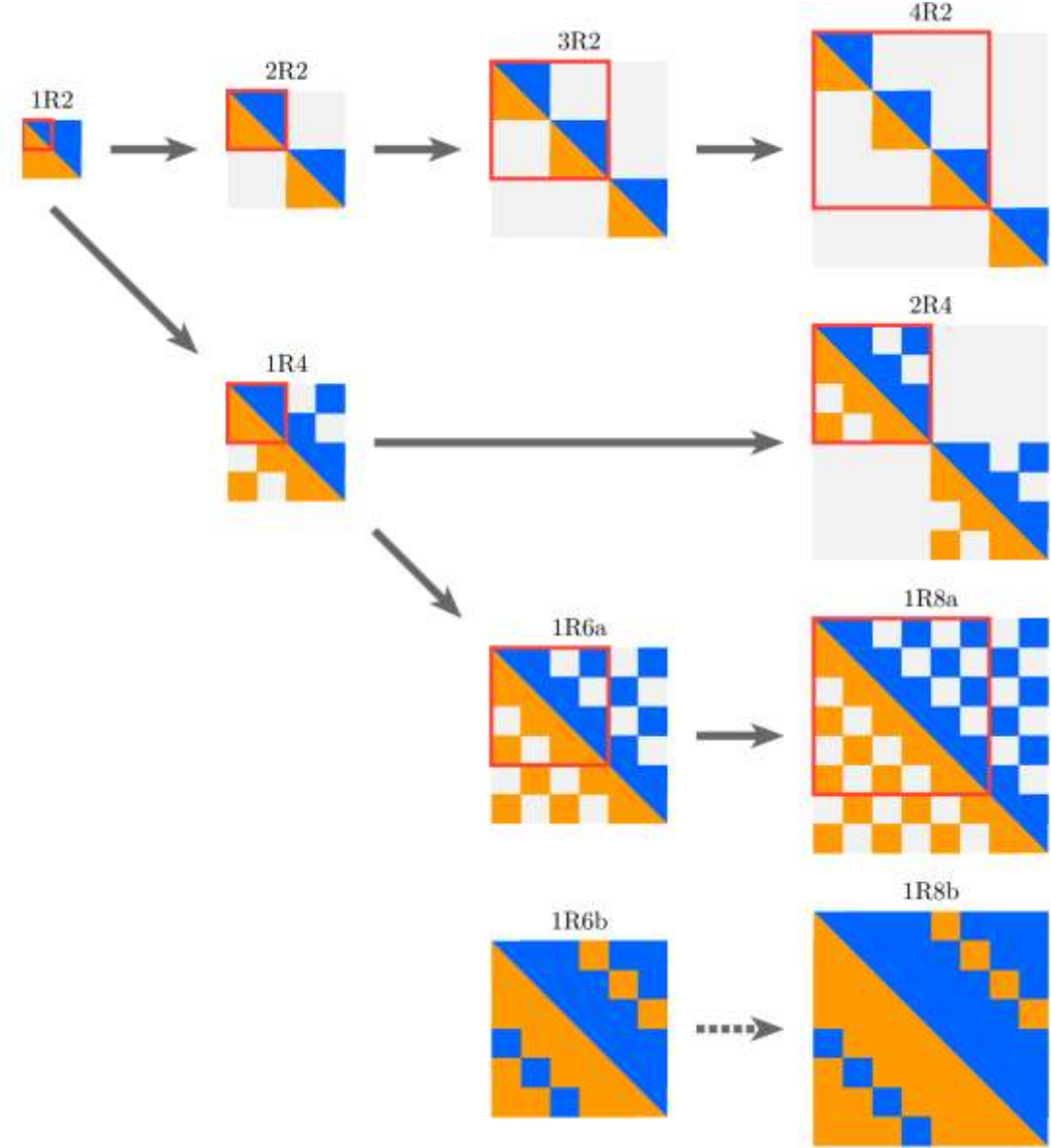
- We evaluate the path integral at large N by finding the leading saddles, but the numerics are demanding (replicas, broken time-translation, ...)

$$\mathcal{I}_{\text{SYK}}^{(R,w)}[G, \Sigma] = - (1 - w) \ln \text{pf}[I_R \otimes \mathcal{D}^- - \Sigma] - w \ln \text{pf}[I_R \otimes \mathcal{D}^+ - \Sigma] \\ + \frac{1}{2} \sum_{r,s=1}^R \int_0^\beta d\tau d\tau' \left(\Sigma_{rs}(\tau, \tau') G_{rs}(\tau, \tau') - \frac{J^2}{q} G_{rs}(\tau, \tau')^q \right)$$

$$\overline{\xi(w)^R} \approx \exp \left(-N \left[\mathcal{I}_{\text{SYK}}^{(R,w)} - R \mathcal{I}_{\text{SYK}}^{(1,0)} \right] \right)$$

$$\mathcal{D}^\mp = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & \pm 1 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{pmatrix}$$

R	Structure	G_1	G_2	G_3	G_4	G_5	G_6	G_7	Comments
2	$\mathbf{R}_2(+)$	$1/2$							
4	$\mathbf{R}_4(0,+)$	0	$1/2$	0					$\cong 2 \times \mathbf{R}_2(+)$
	$\mathbf{R}_4(+,0)$	$1/2$	0	$1/2$					
	$\mathbf{R}_4(+,+)$	$1/2$	$1/2$	$1/2$					$\rightsquigarrow \mathbf{R}_4(+,0)$ limit. convergence
6	$\mathbf{R}_6(0,0,+)$	0	0	$1/2$	0	0			$\cong 3 \times \mathbf{R}_2(+)$
	$\mathbf{R}_6(0,+,0)$	0	$1/2$	0	$1/2$	0			limit. convergence
	$\mathbf{R}_6(0,+,+)$	0	$1/2$	$1/2$	$1/2$	0			$\rightsquigarrow \mathbf{R}_6(+,+, -)$
	$\mathbf{R}_6(+,0,0)$	$1/2$	0	0	0	$1/2$			$\rightsquigarrow \mathbf{R}_6(+,0,+)$
	$\mathbf{R}_6(+,0,+)$	$1/2$	0	$1/2$	0	$1/2$			
	$\mathbf{R}_6(+,0,-)$	$1/2$	0	$-1/2$	0	$1/2$			limit. convergence
	$\mathbf{R}_6(+,+,0)$	$1/2$	$1/2$	0	$1/2$	$1/2$			$\rightsquigarrow \mathbf{R}_6(+,+, -)$
	$\mathbf{R}_6(+,+,+)$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$			$\rightsquigarrow \mathbf{R}_6(+,0,+)$
	$\mathbf{R}_6(+,+, -)$	$1/2$	$1/2$	$-1/2$	$1/2$	$1/2$			
8	$\mathbf{R}_8(0,0,0,+)$	0	0	0	$1/2$	0	0	0	$\cong 4 \times \mathbf{R}_2(+)$
	$\mathbf{R}_8(0,0,+,0)$	0	0	$1/2$	0	$1/2$	0	0	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(0,0,+,+)$	0	0	$1/2$	$1/2$	$1/2$	0	0	$\rightsquigarrow \mathbf{R}_8(+,-,+,+)$
	$\mathbf{R}_8(0,+,0,0)$	0	$1/2$	0	0	0	$1/2$	0	$\cong 2 \times \mathbf{R}_4(+,0)$
	$\mathbf{R}_8(0,+,0,+)$	0	$1/2$	0	$1/2$	0	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(0,+,0,0)$
	$\mathbf{R}_8(0,+,+,0)$	0	$1/2$	$1/2$	0	$1/2$	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(0,+,+,+)$	0	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(0,+,+,-)$	0	$1/2$	$1/2$	$-1/2$	$1/2$	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(+,0,+,0)$	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	
	$\mathbf{R}_8(+,0,+,+)$	$1/2$	0	$1/2$	$1/2$	$1/2$	0	$1/2$	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(+,+,+,0)$	$1/2$	$1/2$	$1/2$	0	$1/2$	$1/2$	$1/2$	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(+,+,+,+)$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(+,+,+,-)$	$1/2$	$1/2$	$1/2$	$-1/2$	$1/2$	$1/2$	$1/2$	



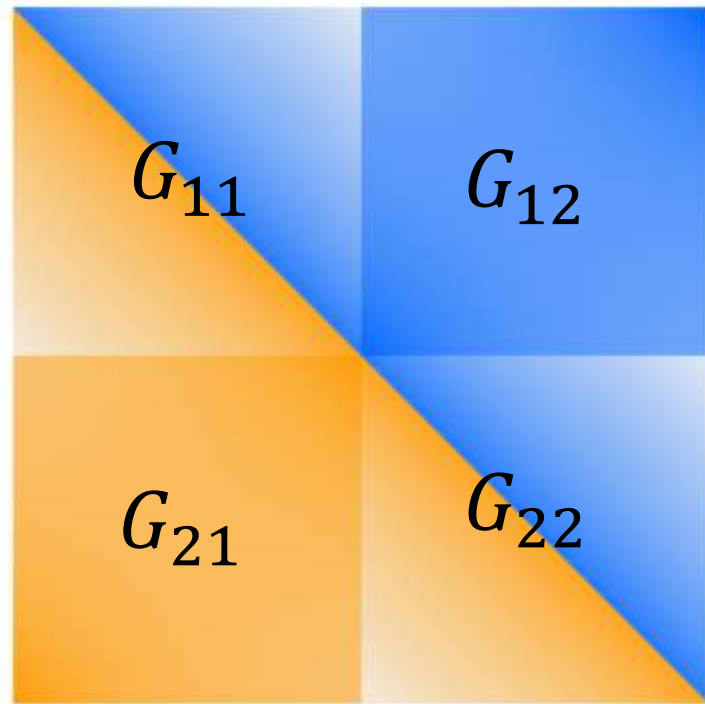
Path integral results

R=4



- We find the 1-point functions are **real gaussian** at large N provided $q > 2$ (SYK interaction degree)

R=2

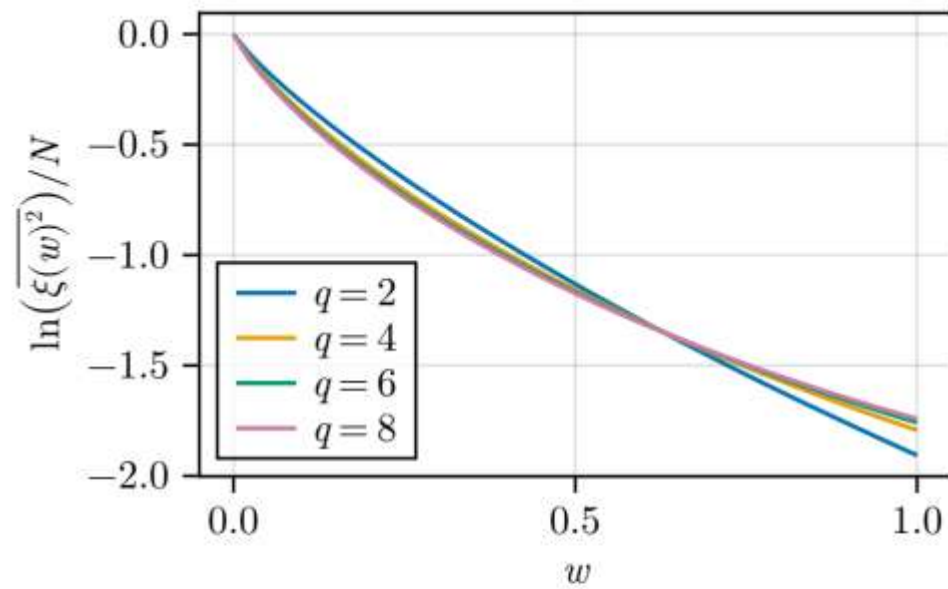


(b) $\beta = 5, w = 0.5$

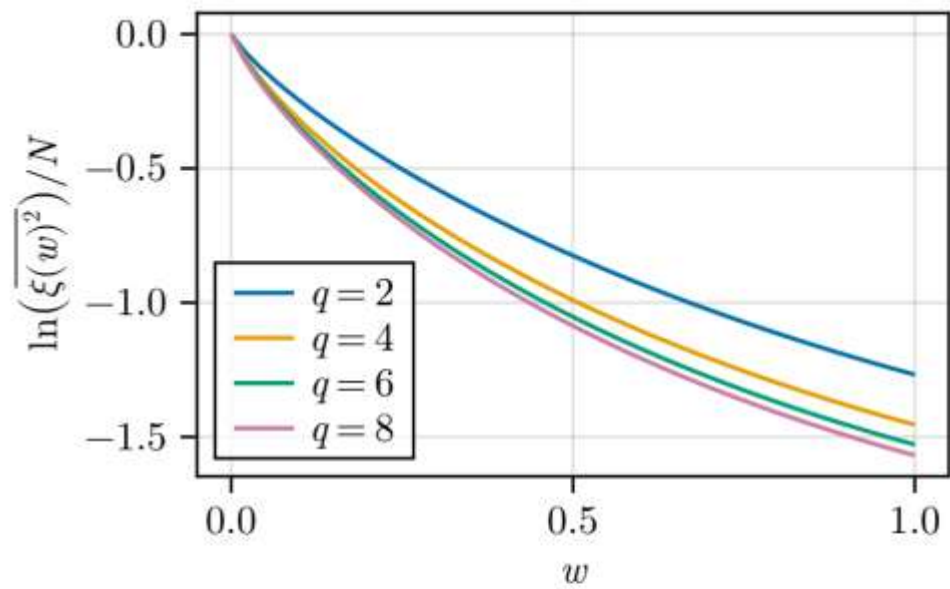


(e) $\beta = 20, w = 0.5$

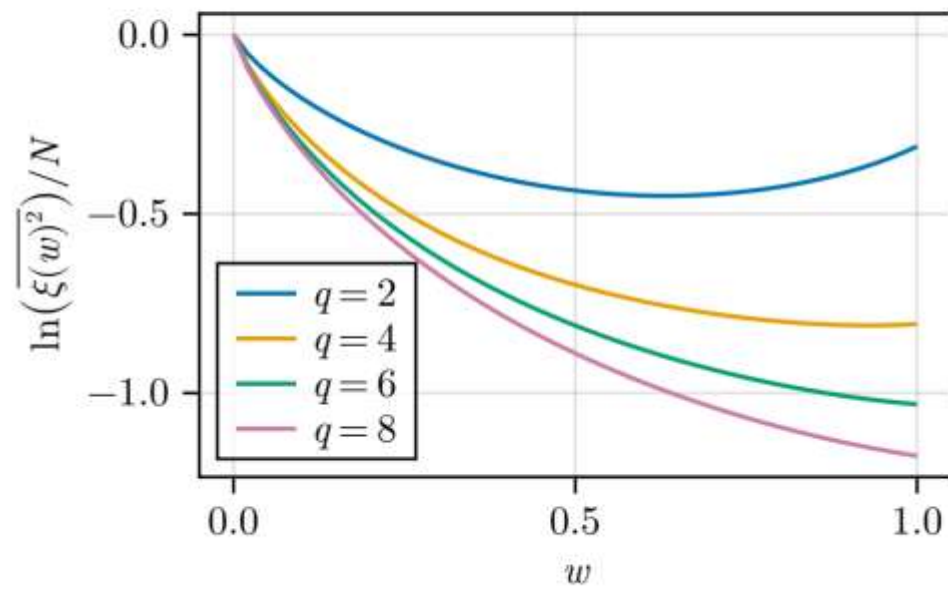
[Bettaque-S]



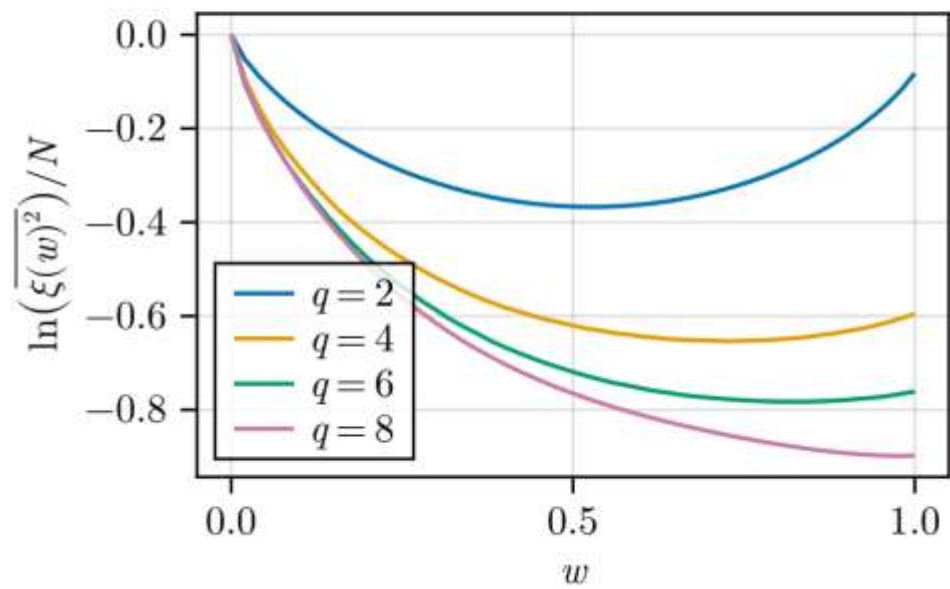
(a) $\beta = 0.5$



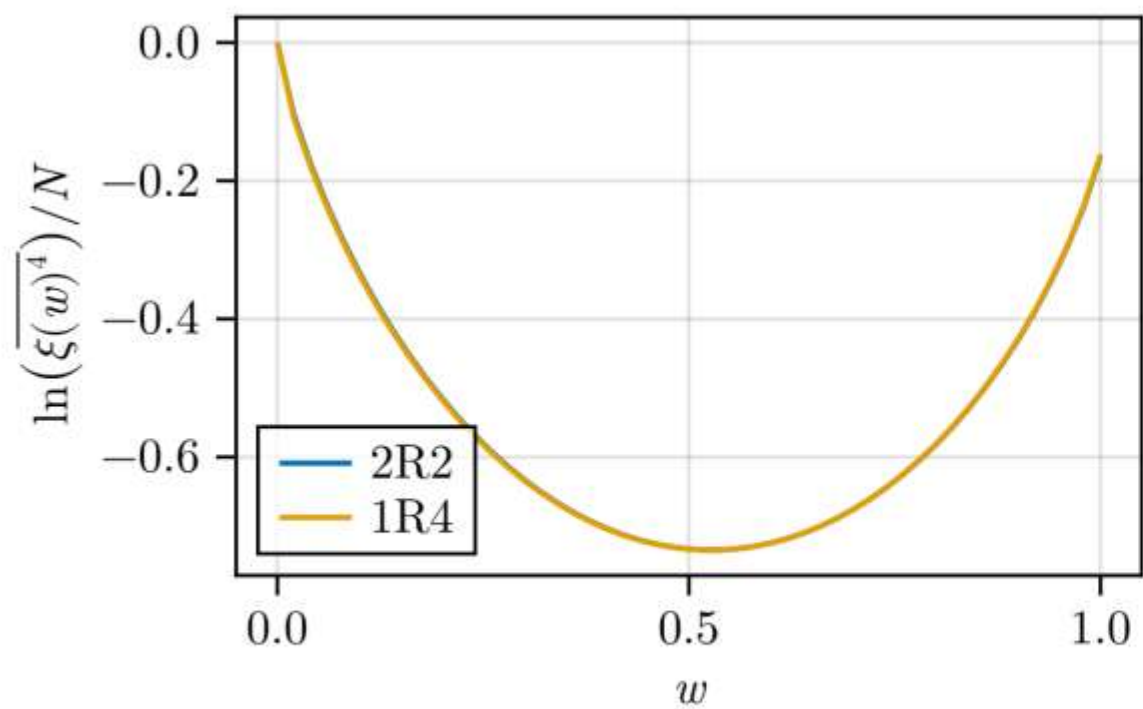
(b) $\beta = 1$



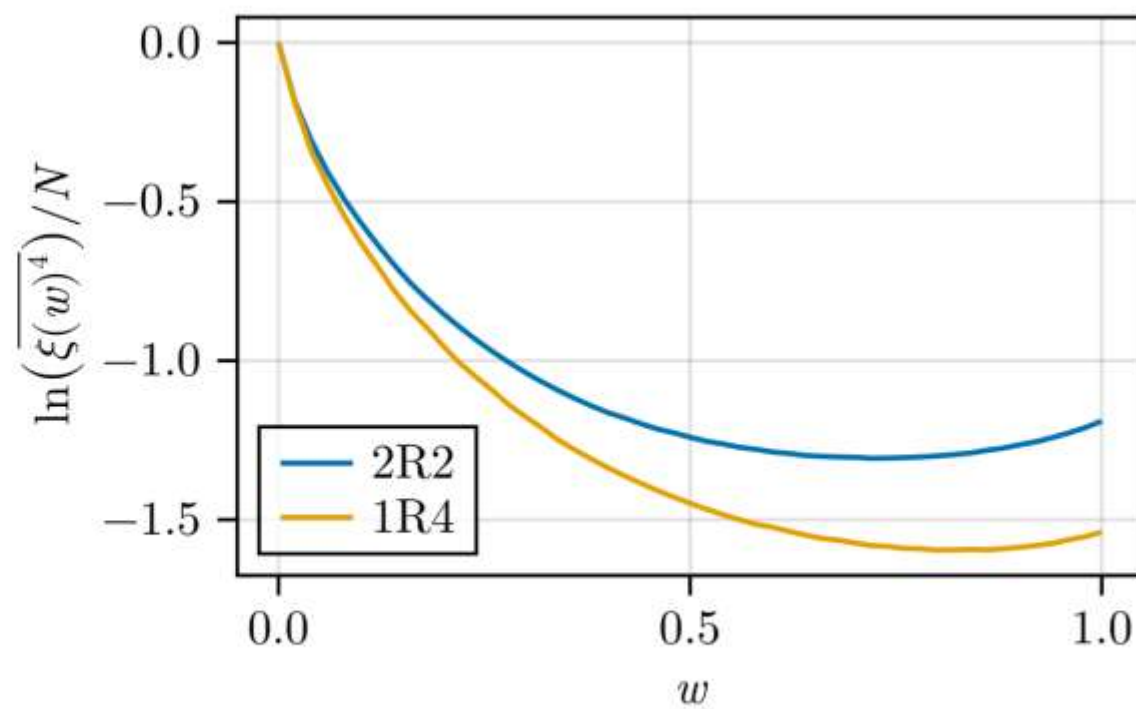
(c) $\beta = 5$



(d) $\beta = 20$



(a) $q = 2$

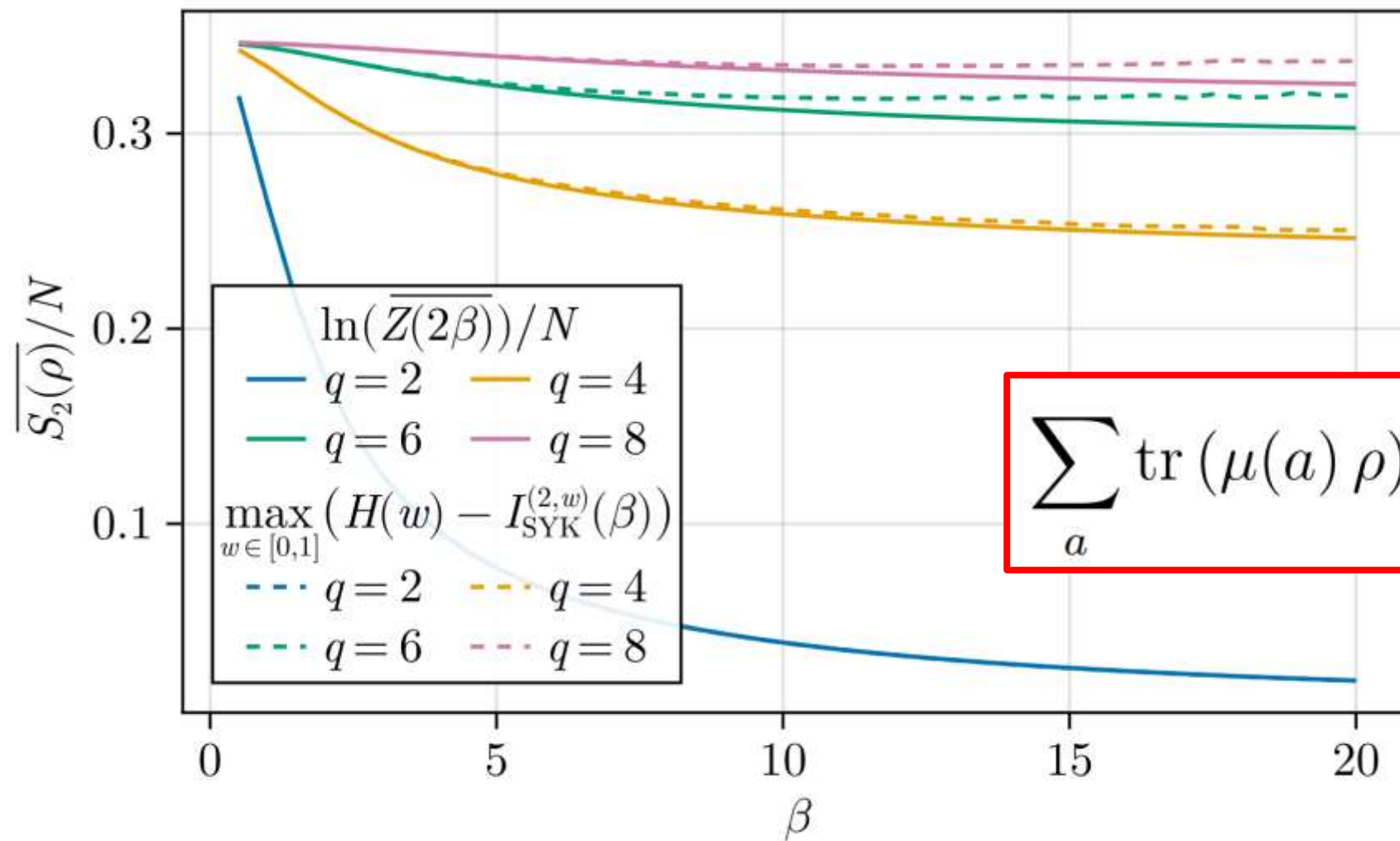


(b) $q = 4$

enhanced $O(R)$ replica symmetry

R=2 and purity

$$\overline{\sum_a \text{tr} (\mu(a) e^{-\beta H_{\text{SYK}}})^R} \approx \int_0^1 dw \exp \left(N \left[H(w) - \mathcal{I}_{\text{SYK}}^{(R,w)} \right] \right)$$

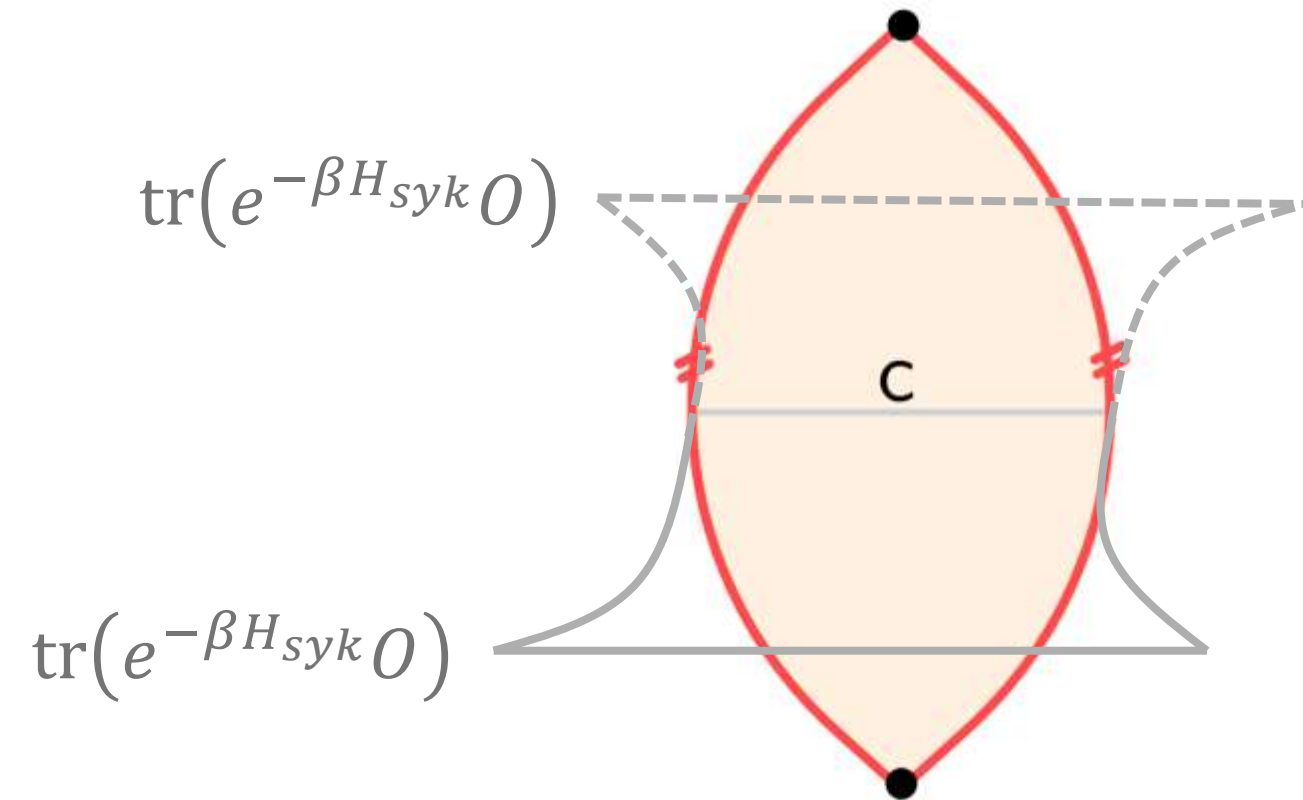


$$\sum_a \text{tr} (\mu(a) \rho)^2 = D \text{tr}(\rho^2)$$

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- Magic in SYK
- Advantage in SYK?
- Generalizing, e.g. Hubbard?

Enter wormholes and closed universes



We model the inter-replica correlations, G_{21} , as correlations between the two sides of a Euclidean wormhole; analytic continuation gives a closed universe [Maldcena-Maoz]

[JT wormhole/cosmology solutions: Stanford, Usatyuk-Wang-Zhao]

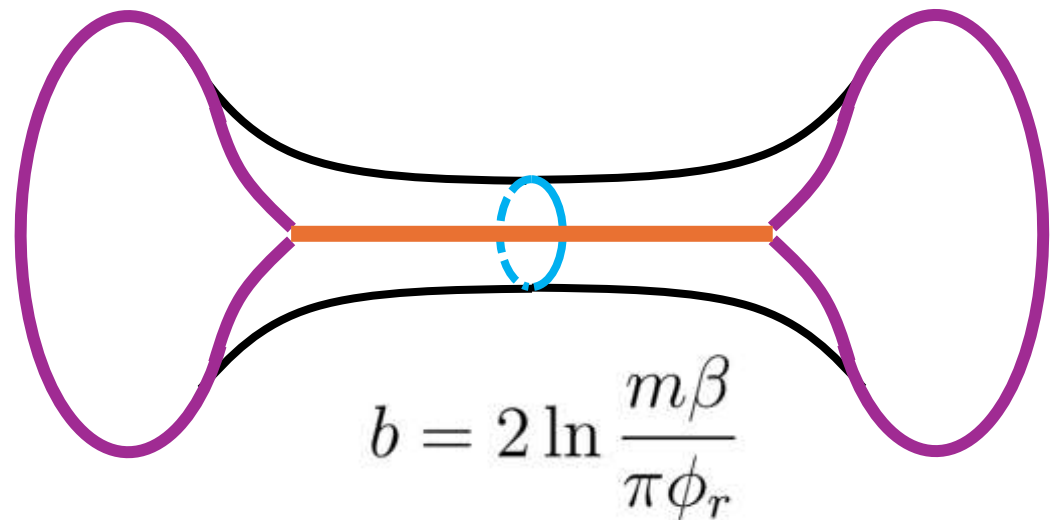
Wormhole calculation

- We model the SYK inter-replica correlation with a boundary-to-boundary correlations of N light probe fermions moving in a background geometry stabilized by a heavy particle [Bettaque-S]
- The bulk 2-point function of the light fermions is computed by summing over geodesics with mass $\rightarrow$ scaling dimension
- Background – double trumpet

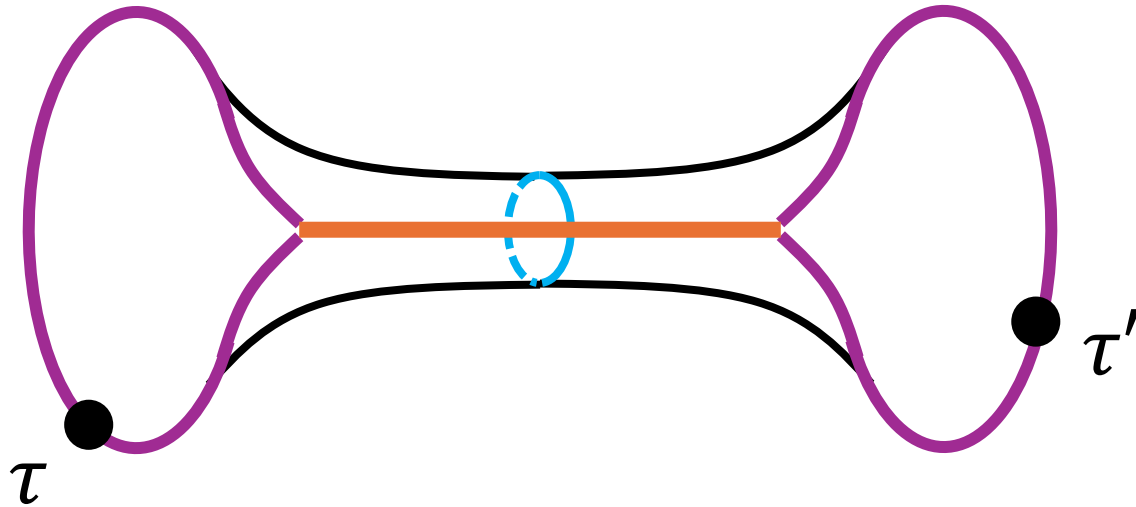
$$ds^2 = d\rho^2 + b^2 \cosh^2 \rho d\sigma^2$$

$$\Phi = \frac{\pi\phi_r}{\beta} \cosh \rho \cosh b\sigma.$$

[Stanford, Usatyuk-Wang-Zhao]



Wormhole calculation



$$e^{\rho_c} = \frac{2\beta}{\pi\epsilon} \frac{1}{\cosh b\sigma}.$$

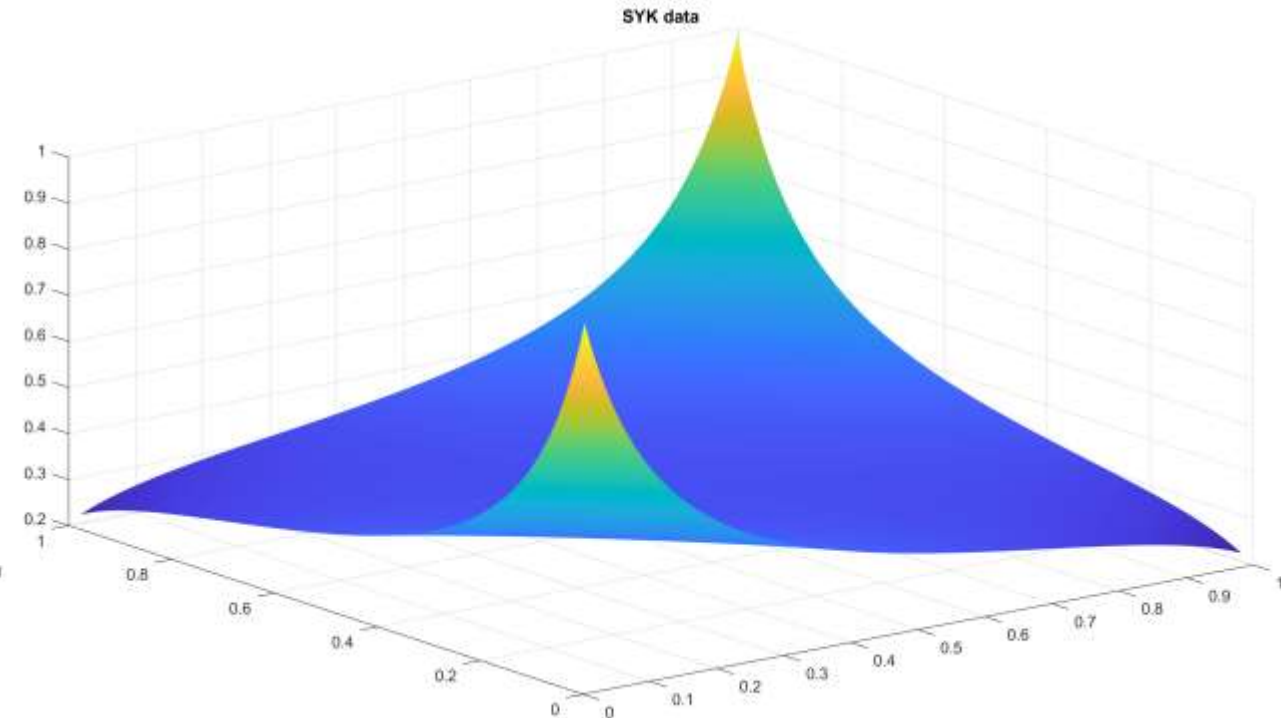
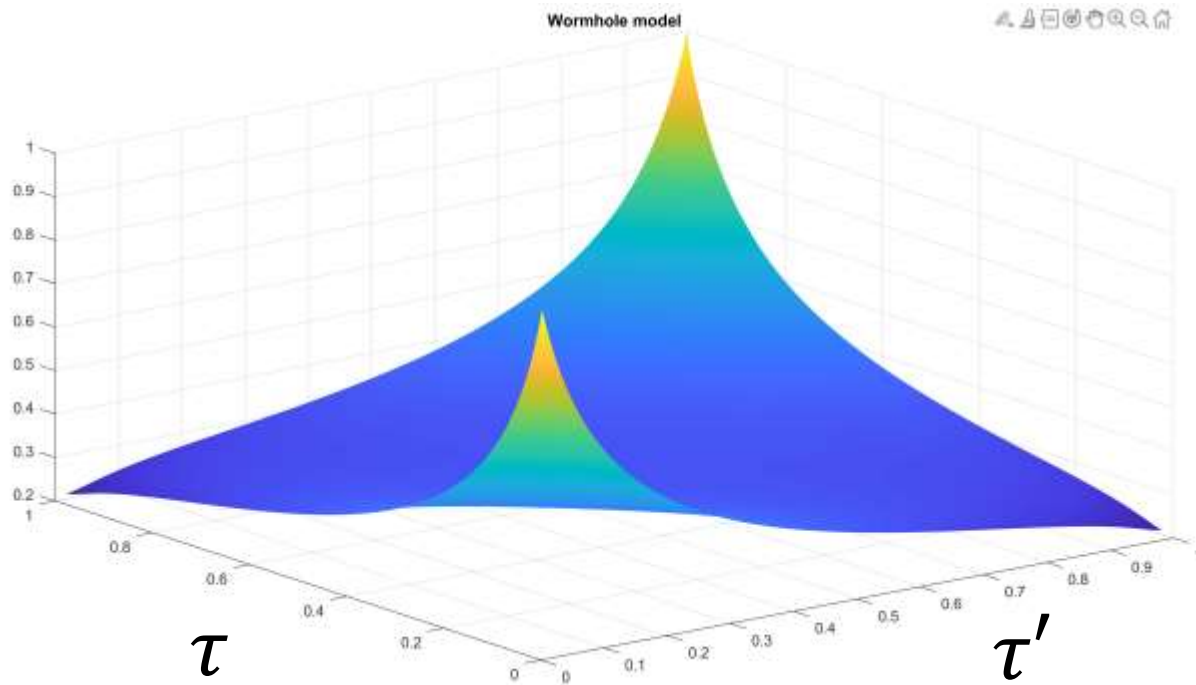
$$\frac{\tau(\sigma)}{\beta} = \frac{\arctan e^{b\sigma} - \arctan e^{-b/2}}{\arctan e^{b/2} - \arctan e^{-b/2}}.$$

winding geodesics $d_n(\sigma, \sigma') = \rho_c(\sigma) + \rho_c(\sigma') + \log \cosh^2 \frac{b(\sigma - \sigma' - n)}{2}.$

$$\mathcal{G}_P(\tau, \tau') = \mathcal{N}_P \sum_n e^{-\Delta d_n(\sigma(\tau), \sigma'(\tau'))}, \quad \mathcal{G}_{AP}(\tau, \tau') = \mathcal{N}_{AP} \sum_n (-1)^n e^{-\Delta d_n(\sigma(\tau), \sigma'(\tau'))},$$

Comparison of normalized correlations

$$\mathcal{G}_f = (1 - f)\mathcal{G}_{AP} + f\mathcal{G}_P.$$



Bulk model: $b = 6.4, f = .32$; SYK: $w = .25, \beta J = 20, L = 1000$

[[Bettaque-S](#), larger beta WIP [Pollack-Milgate-S](#)]

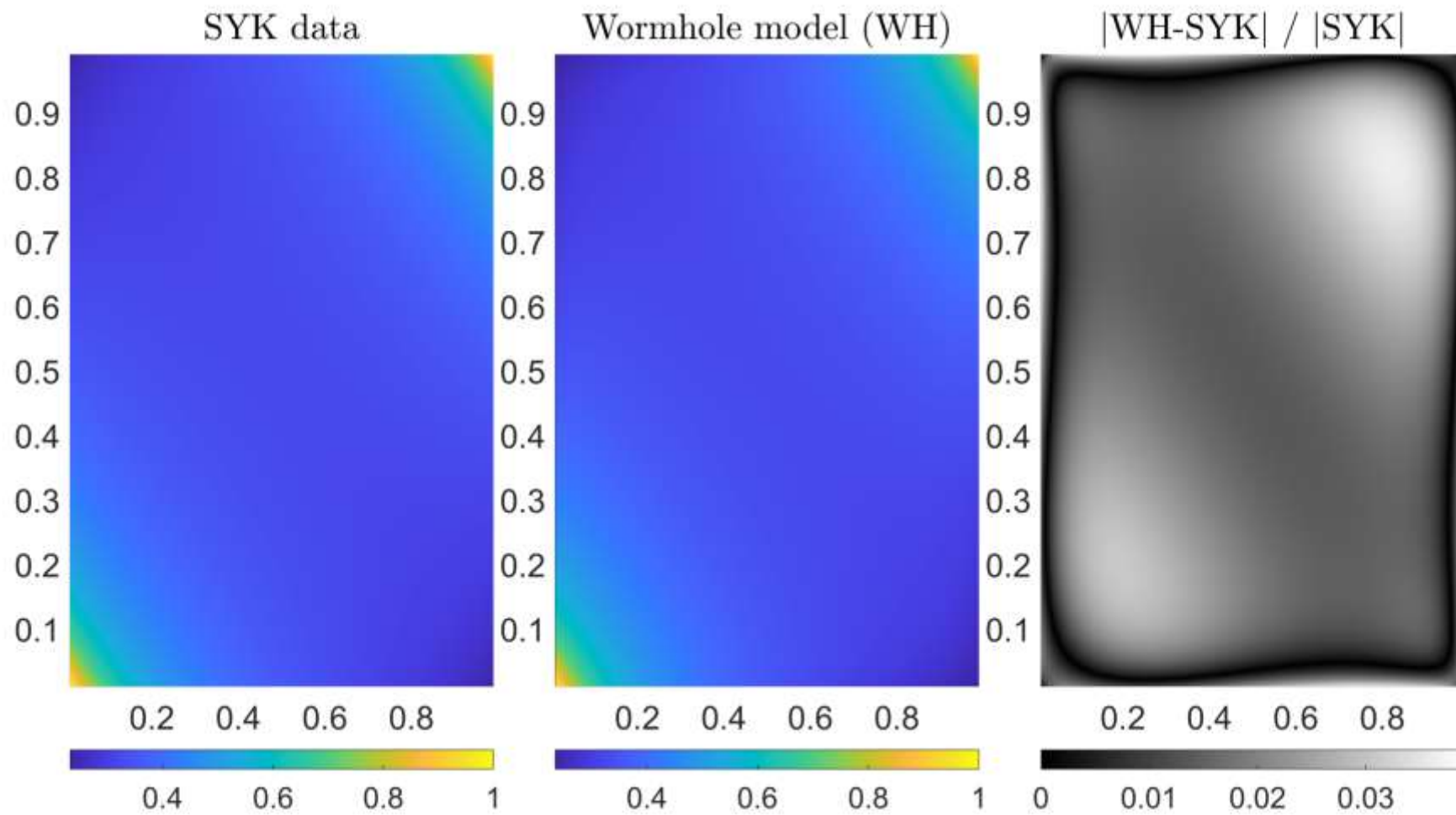
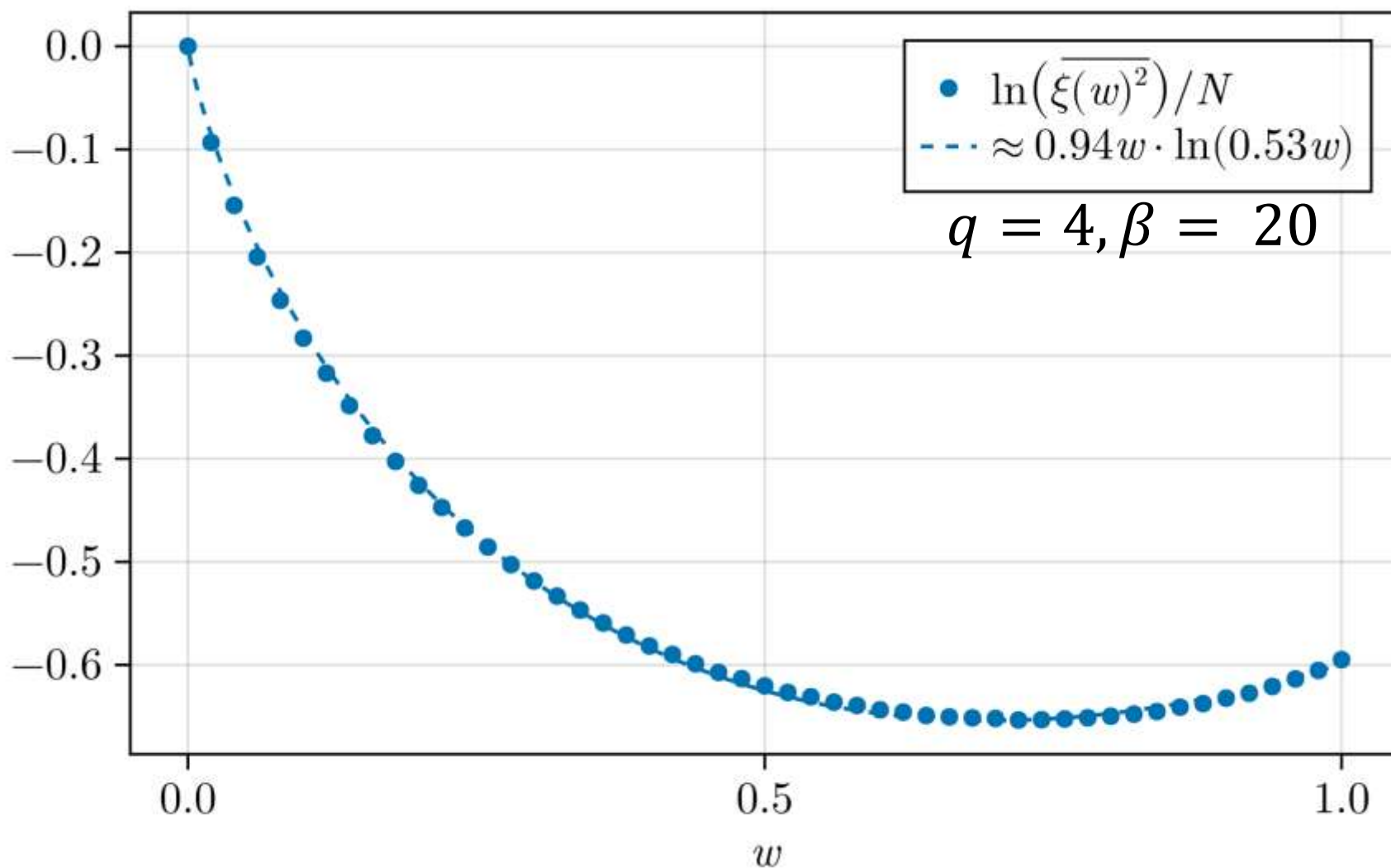


Figure 21: (Left) The plot shows the value of $G_{21}(\tau, \tau')$ in color as a function of τ/β and τ'/β . It is normalized by its maximum value, which occurs in the corners of the square. The SYK parameters are $\beta = 20$, $w = .25$, $L = 2000$. (Middle) The same configuration now showing the normalized wormhole prediction obtained by fitting to the SYK data. The fit values are $b = 6.5$ and $f = .33$. (Right) The absolute value of the relative difference between (Left) and (Middle). The maximum deviation is around 4%.

Action data

$$I_{\text{JT}} = -S_0\chi - \frac{1}{2} \int d^2x \sqrt{g} \Phi (R + 2) + (\text{boundary terms}) \textcolor{red}{- m\ell_H} + I_{\text{matter}}$$



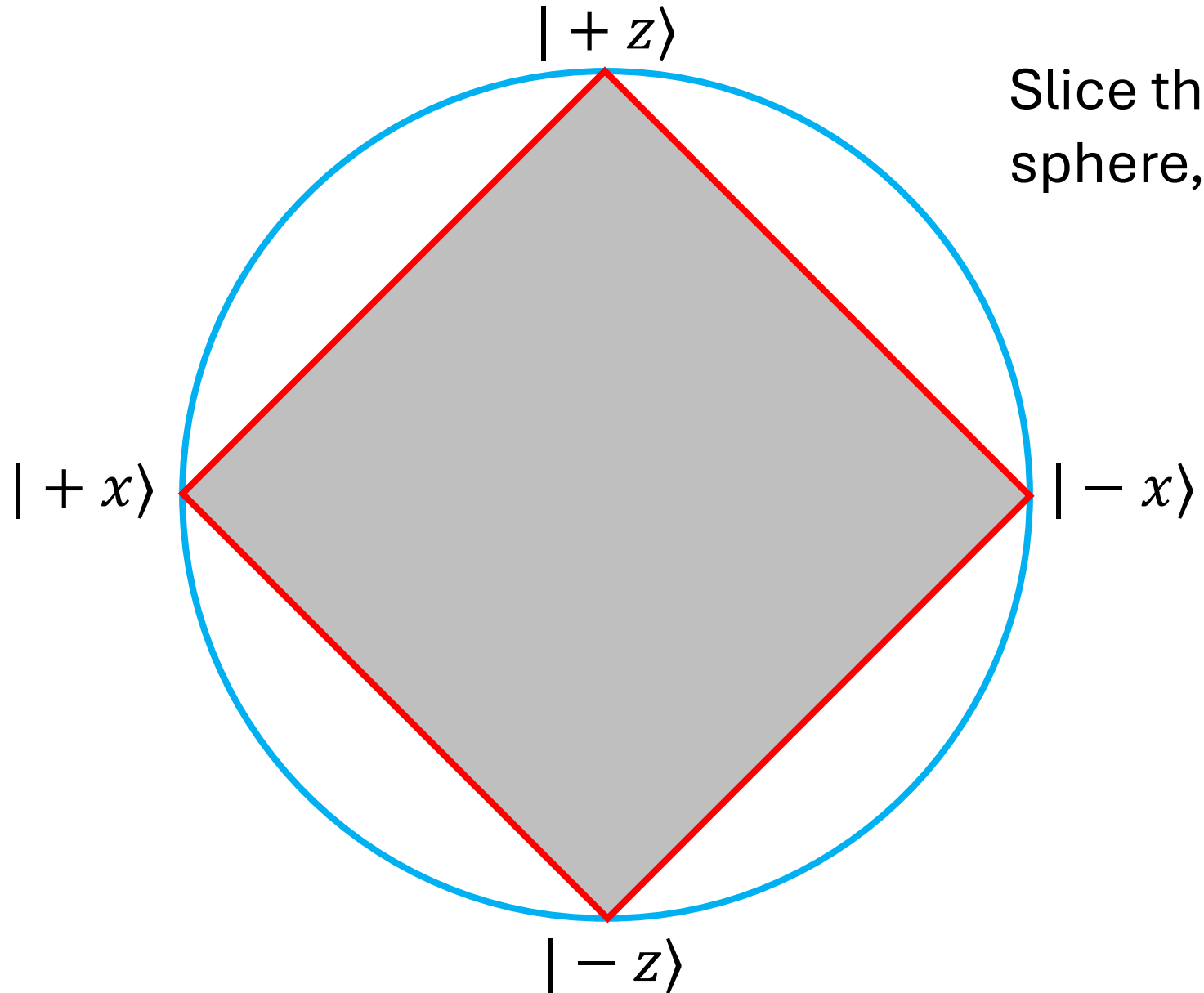
Plan

- Setting the stage
 - Motivation from wormholes and closed universes
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- Generalizing, e.g. Hubbard?

Resources in quantum theory

- One useful way to view physical theories is in terms of resources, e.g. quantum entanglement
 - A. “Free states” = unentangled states
 - B. Resource measures = entanglement entropy, ...
- Entanglement is necessary for teleportation, multi-player quantum games, and it is related to the geometry of spacetime
- From the perspective of classical simulation, low entanglement offers an opportunity that we can take advantage
- Magic is the resource theory of “non-stabilizerness”

Single qubit **stabilizer polytope** (STAB = mixtures of stabilizer states):



Slice through the Bloch sphere, **x-z plane**

Fermionic Cliffords and Stabilizers

Clifford group maps strings
to strings, no superposition: $\mu(a) \rightarrow U\mu(a)U^\dagger \propto \mu(a')$

Stabilizer states are obtained by acting on a fixed product basis
($2i\chi_{2j-1}\chi_{2j} = 1$) with Cliffords, which imparts classical simulability

Recently, we proved a number of facts about this group, which turns out to be $O(N, Z_2)$ after demanding fermion parity symmetry; we also showed that it forms 3-design and gave an efficient sampling algorithm

[Bettaque-S 2407.11319]

Measures of magic

- There are many measures of magic, which tend to involve analogues of distance to the stabilizer polytope (STAB)
- Many of these are not efficiently computable, and the situation is also harder when dealing with mixed states
- We will focus for simplicity on one measure, **robustness of magic**

$$R(\rho) = \min \left(\sum_{\alpha} |c_{\alpha}| : \rho = \sum_{\alpha} c_{\alpha} \sigma_{\alpha}, \sigma_{\alpha} \in \text{STAB} \right)$$

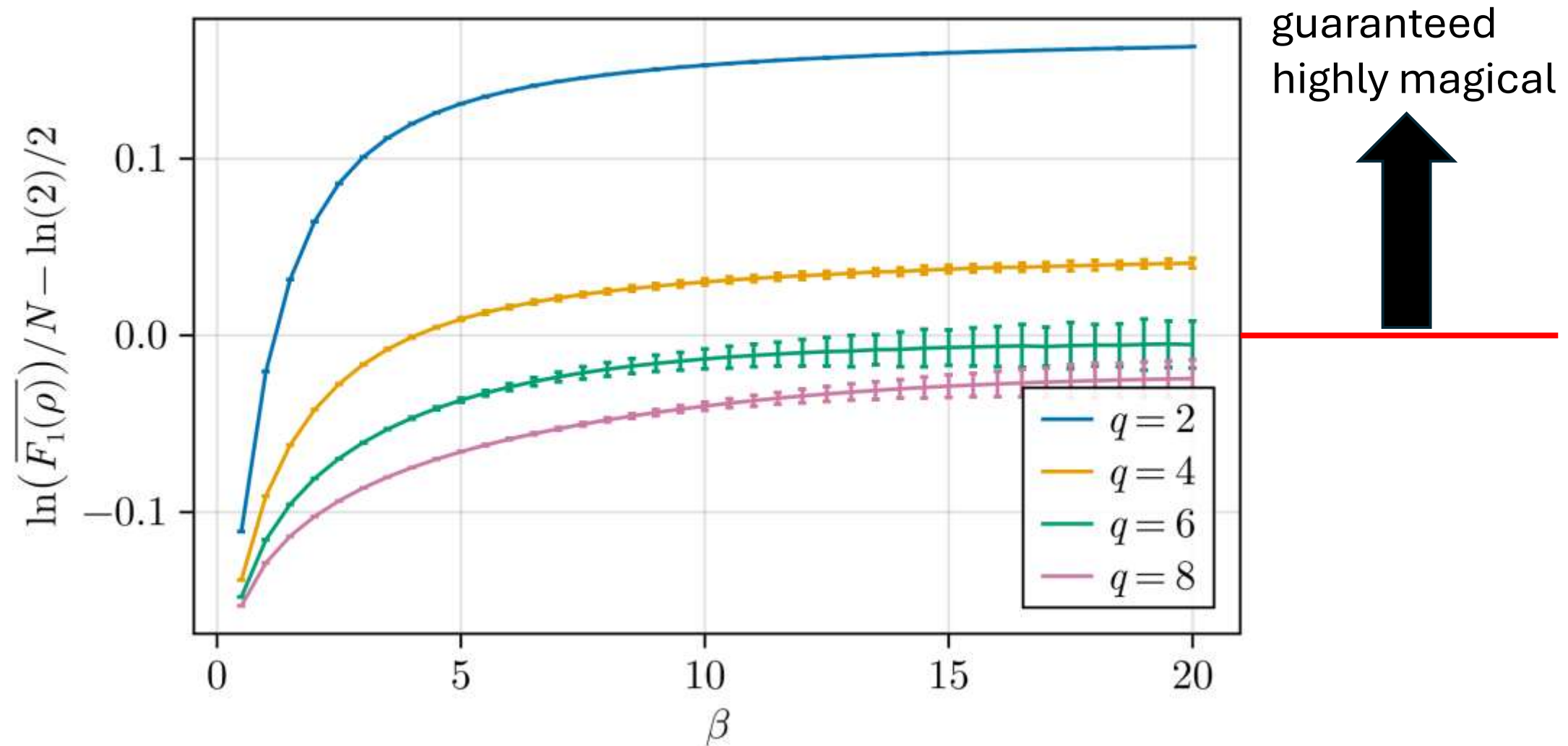
$$R(\rho) \geq \frac{F_1(\rho)}{D}, F_1(\rho) = \sum_a |\xi(a)|$$

Many-body magic

- The idea of magic is relatively old, going back to [Bravyi-Kitaev](#), and is well-studied in the quantum information community
- In 2020, many-body magic was studied for the first time in the 3-state Potts model [[Cao-S-White](#)] and beyond [[Liu-Winter](#)]
- There is now an explosion of interest, including the development of novel measures such as stabilizer Renyi entropy [[Leone-Oliviero-Hamma](#), ...], magic in SYK [[Bera-Schiro](#), [Jasser-Odavic-Hamma](#), [Zhang-Zhou-Sun](#), ...], and proposed connections to gravitational backreaction in holography [[Cao-Cheng-Hamma-Leone-Munizzi-Oliviero](#)]

Robustness of magic

$$R(\rho) \geq \frac{F_1(\rho)}{D}, F_1(\rho) = \sum_a |\xi(a)|$$



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Thermal observables:

$$\langle \dots \rangle = \frac{\text{Tr}(\dots e^{-\beta H})}{\text{Tr}(e^{-\beta H})}$$

Example: observable is zero on average, but small and non-zero in each instance:

$$\langle \psi_i \psi_j \psi_k \psi_l \rangle = ?$$

Is it possible to calculate these efficiently and to high accuracy using only classical resources?

Yes, and the algorithm is essentially dressed perturbation theory



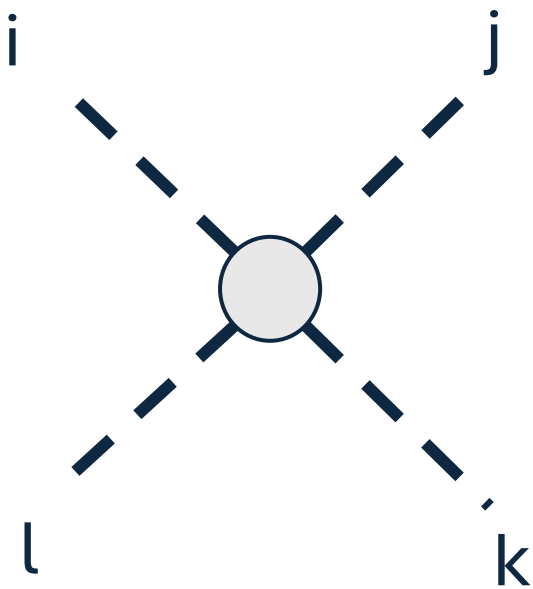
Start with a simple high-temperature expansion:

$$e^{-\beta H} = 1 - \beta H + \underbrace{\frac{1}{2}\beta^2 H^2 + \dots}$$

$$\langle \psi_1 \psi_2 \psi_3 \psi_4 \rangle = 0 - \# \beta J_{1234} + \dots$$

$$\tau \quad \text{-----} \quad \tau'$$

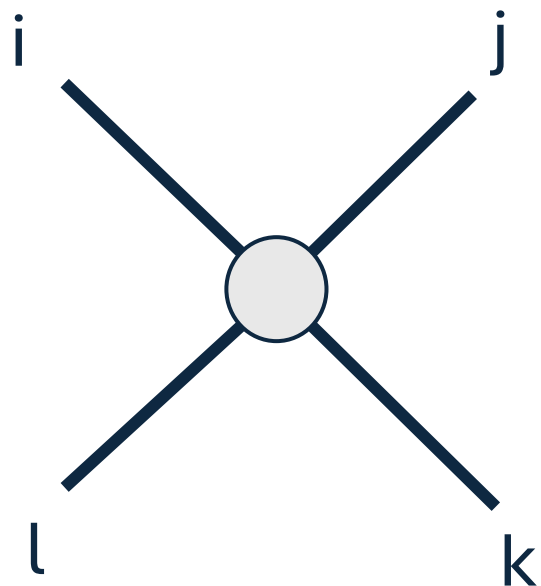
$$G_0(\tau, \tau') = \frac{1}{2} \text{sgn}(\tau - \tau')$$



$$-J_{ijkl} \int d\tau G_0(\tau, 0)^4 = -\frac{\beta}{16} J_{1234}$$

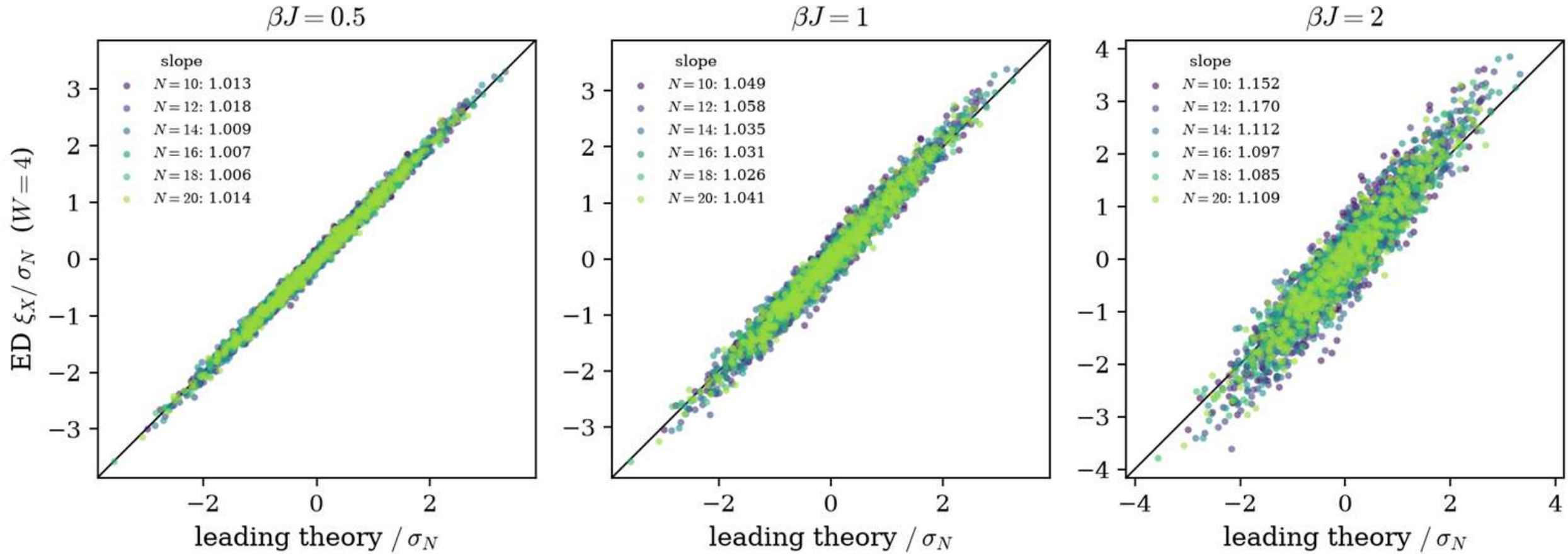
Melonic dressing:

$$\overline{G(\tau, \tau')} = \text{---} + \text{---} \circ \text{---} + \dots$$



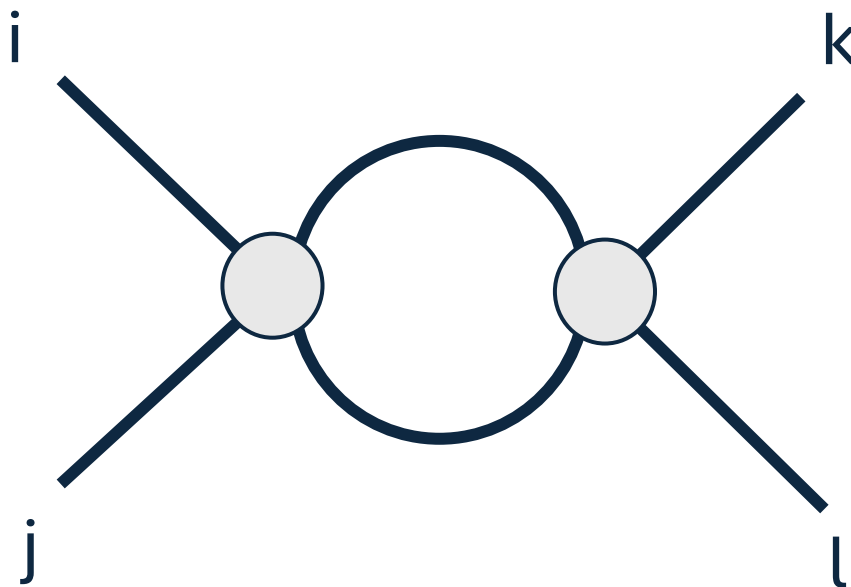
$$-J_{ijkl} \int d\tau G(\tau, 0)^4$$

Leading diagram does well, but degrades as β is increased:



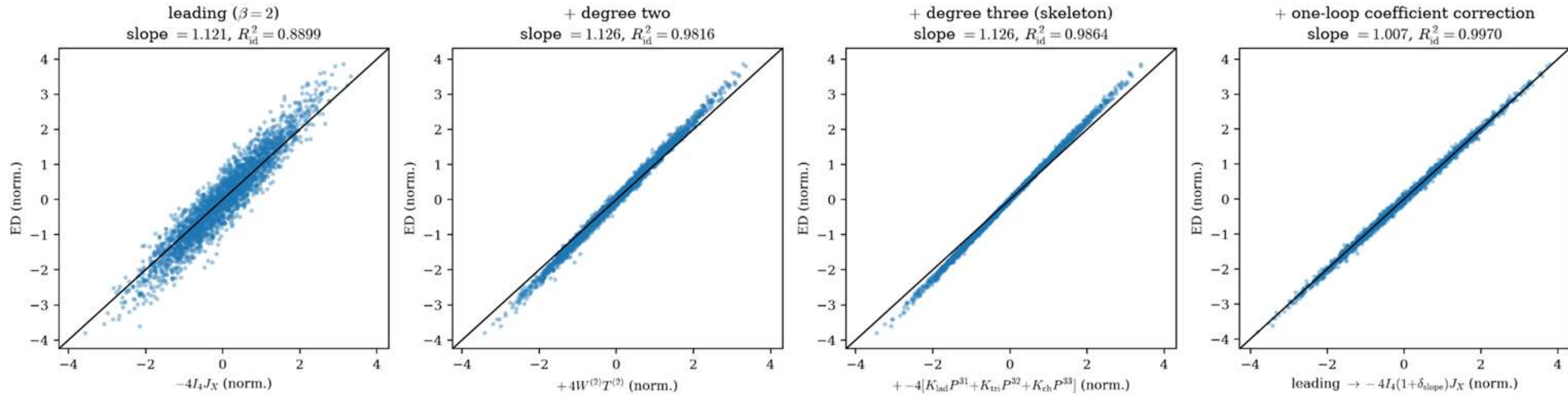
Add corrections:

$$\sum_{uv} J_{ijuv} J_{uvkl}$$

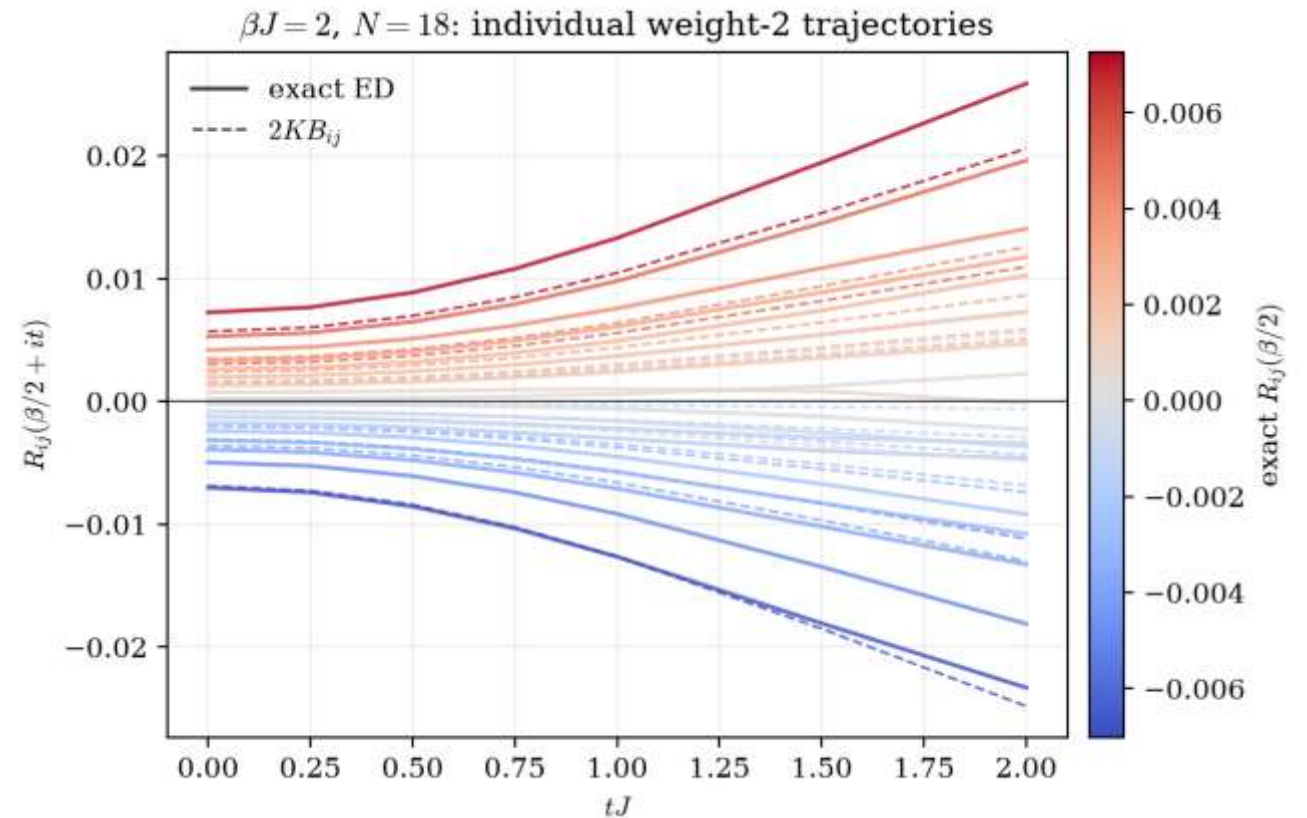
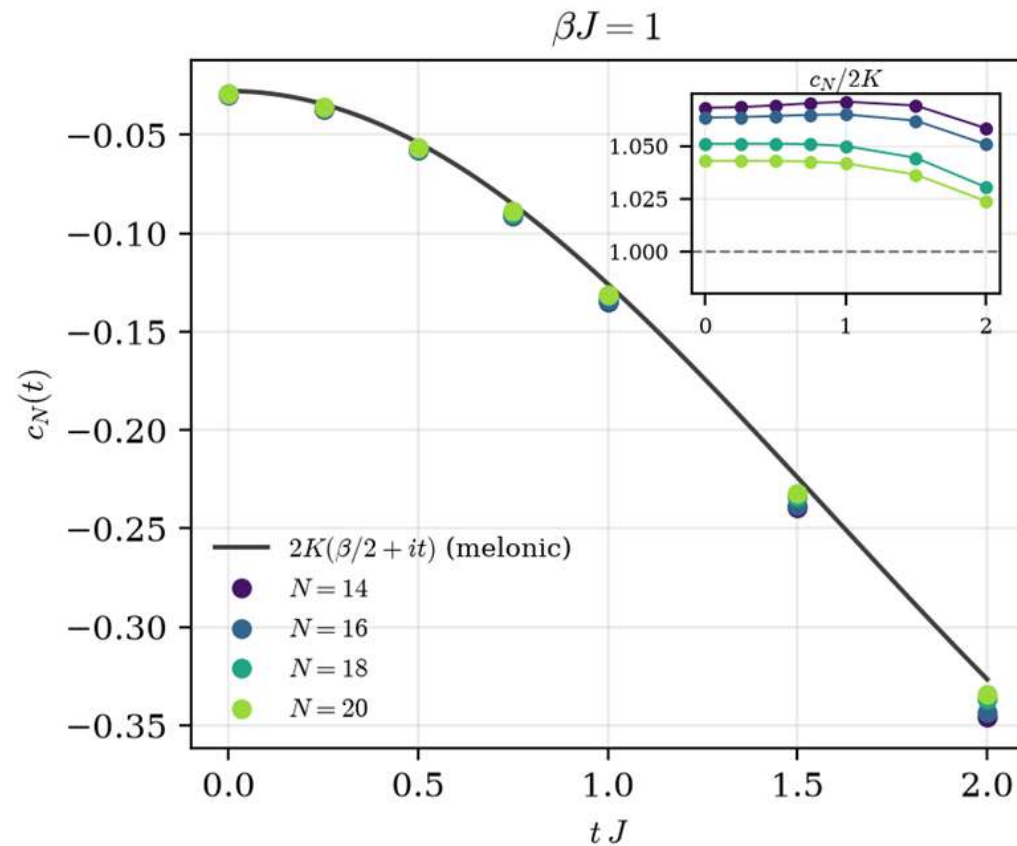


$$\int d\tau_1 d\tau_2 G(\tau_1)^2 G(\tau_1 - \tau_2)^2 G(\tau_2)^2$$

Going to third order in the expansion:



We can even extend the method to real time dynamics:



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What about local models?

- What if we care about a more realistic model of strongly correlated electrons, e.g. the Hubbard model? Not clear yet, but we have some preliminary results [Guo-S WIP]
- Toy model: weight W Pauli string in a random product state $\rightarrow$ 1-point function has a random sign and **log-normal magnitude**
- **Log-normal, rather than normal, seems to be the norm for many local models, e.g. finite depth circuits, MPS, critical-but-free theories**
- So perhaps the Hubbard model doesn't have the right structure to match a simple wormhole picture of its 1-point statistics? Or maybe if interactions are strong enough or we use the right ops?

Outlook

- Several motivations {closed universes, magic, quantum advantage?} converged on the relevance of the statistics of multi-body thermal 1-point functions
- So far, we've identified two broad classes: normal and log-normal, but this is an underexplored area distinct from ETH!
- Many questions: Can we set up a table-top experiment to holographically probe a singularity?, Can we leverage these magic results to devise better classical simulation methods?, Can we like-wise more deeply understand what makes SYK so solvable and the limits thereof?, ...